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COMEDK 2025 Question Paper With Solution

Consortium of Medical, Engineering and Dental
Colleges of Karnataka Undergraduate
Entrance Test (COMEDK UGET)

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COMEDK 2025 Question Paper

Max. Marks: 180

Duration: 3 Hours

1. This paper consists of 180 questions with 3 parts of Chemistry, Mathematics and Physics
 - **Chemistry:** (Q. No. 1 to 60) Multiple Choice Questions with one correct answer. A correct answer carries 1 Mark. No Negative marks.
 - **Mathematics:** (Q. No. 61 to 120) Multiple Choice Questions with one correct answer. A correct answer carries 1 Mark. No Negative marks.
 - **Physics:** (Q. No. 121 to 180) Multiple Choice Questions with one correct answer. A correct answer carries 1 Mark. No Negative marks.
2. The OMR sheet for 200 questions is to be used
3. Use of calculators and log tables is prohibited
4. Darken the appropriate bubble using a pen in the OMR sheet provided to you. Once entered, the answer cannot be changed. Any corrections or modifications will automatically draw a penalty of 1 mark
5. No clarification will be entertained during the examination. Doubts in the paper can be reported to the coordinator after the exam
6. If the details in the OMR Sheet are not filled, If the OMR sheet is mutilated, torn, white Ink used, the circles filled and scratched, then the OMR sheet will not be graded

All the best!!

Useful Data

At. Wt.:

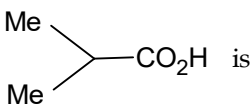
$N = 14$; $O = 16$; $H = 1$; $S = 32$; $Cl = 35.5$; $Mn = 55$; $Na = 23$; $C = 12$; $Ag = 108$; $K = 39$; $Fe = 56$; $Pb = 207$

Physical Constants:

$h = 6.626 \times 10^{-34} \text{ Js}$, $N_a = 6.022 \times 10^{23} \text{ mol}^{-1}$, $c = 2.998 \times 10^8 \text{ ms}^{-1}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

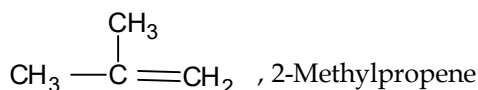
Chemistry

Multiple Choice Questions with one correct answer. A correct answer carries 1 mark. No negative mark. **60 x 1 = 60**

1. Monomer of  is

(a) 2-methylpropene (b) styrene (c) propylene (d) ethene

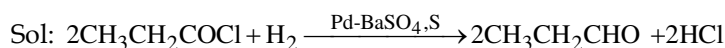
Sol:



Ans: (a)

2. Which of the following reagents will convert $\text{CH}_3\text{CH}_2\text{COCl}$ into $\text{CH}_3\text{CH}_2\text{CHO}$?

(a) LiAlH_4 (b) H_2 / Pd (c) $\text{H}_2 / \text{Pd} - \text{BaSO}_4, \text{S}$ (d) $\text{Na} / \text{C}_2\text{H}_5\text{OH}$



Ans: (c)

3. The energy of an electron in an atomic orbital of a multi-electron atom depends upon

(a) the principal quantum number only
 (b) the principal and azimuthal quantum numbers only
 (c) the principal, azimuthal and magnetic quantum numbers only
 (d) the principal, azimuthal, magnetic and spin quantum numbers.

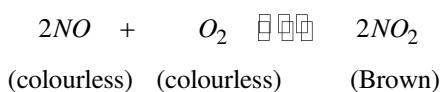
Sol: Energy of electron in multi-electron atom depends upon n and l

Ans: (b)

4. A deep brown gas is formed by mixing two colourless gases. The two colourless gases are

(a) NO_2 and O_2 (b) NO and O_2 (c) N_2O and NO (d) NH_3 and HCl

Sol:

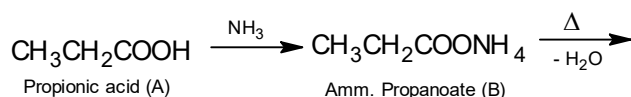


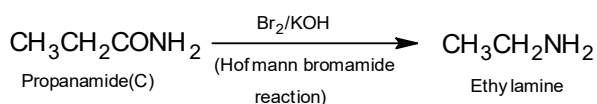
Ans: (b)

5. An organic compound 'A' on treatment with NH_3 gives 'B' which on heating gives 'C', 'C' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is

(a) CH_3COOH (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (c) $\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}}\text{COOH}$ (d) $\text{CH}_3\text{CH}_2\text{COOH}$

Sol:



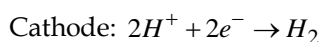
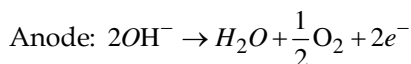


Ans: (d)

6. In the electrolysis of which solution, OH^- ions are discharged in Preference of to Cl^- ions?

- (a) dilute NaCl (b) Very dilute NaCl
(c) fused NaCl (d) Solid NaCl

Sol: In very dilute solution of NaCl , the following takes place.



Ans: (b)

7. Glycogen is a branched chain polymer of α -D-glucose units in which chain is formed by $\text{C}_1 - \text{C}_4$ glycosidic linkage whereas branching occurs by the formation of $\text{C}_1 - \text{C}_6$ glycosidic linkage. Structure of glycogen is similar to

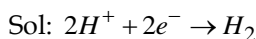
- (a) amylose (b) amylopectin (c) cellulose (d) glucose

Sol: Like glycogen, amylopectin is a branched polymer, amylose and cellulose are linear polymers while glucose is a monomer.

Ans: (b)

8. How many faradays are required to liberate 8 g of hydrogen gas?

- (a) 2 (b) 4 (c) 8 (d) 18



2g H_2 is liberated by 2F of electricity

Ans: (c)

9. The Complex which has the highest magnetic moment among the following is

- (a) $[\text{CoF}_6]^{-3}$ (b) $[\text{Co}(\text{NH}_3)_6]^{+3}$ (c) $[\text{Ni}(\text{NH}_3)_4]^{+2}$ (d) $[\text{Ni}(\text{CN})_4]^{-2}$

Sol: No. of unpaired electrons in

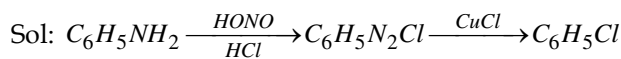
(a) (4)

(d) (c), (b) $\rightarrow$ (0), so $[\text{CoF}_6]^{-3}$ has highest magnetic moment

Ans: (a)

10. Chlorobenzene can be prepared by reacting aniline with

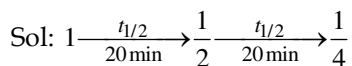
- (a) hydrochloric acid
(b) cuprous chloride
(c) chlorine in presence of anhydrous aluminium chloride
(d) $\text{NaNO}_2 / \text{HCl}$ followed by heating with cuprous chloride



Ans: (d)

11. What fraction of a reactant remains after 40 min if $t_{1/2}$ is 20 min for a first order reaction?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) $\frac{1}{6}$

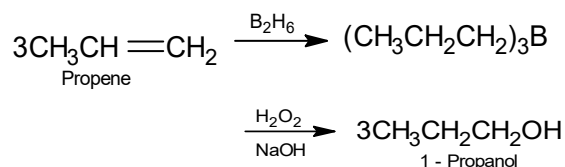


Ans: (a)

12. Propene, $CH_3CH=CH_2$ can be converted into 1-propanol by oxidation. Indicate which set of reagents amongst the following is ideal to effect the above conservation?

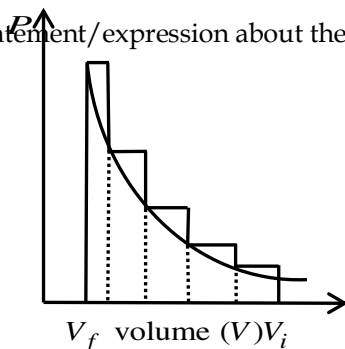
- (a) $KMnO_4$ (alkaline)
(c) B_2H_6 and alk. H_2O_2

Sol: $KMnO_4$ (alkaline) and OsO_4 / CH_2Cl_2 are used for hydroxylation of double bond while O_3 / Zn is used for ozonolysis. Therefore, the right option (c), i.e.,



Ans: (c)

13. Select the correct statement/expression about the graph

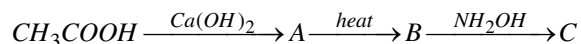


- (a) It is an irreversible compression
(b) Pressure is not constant and changes in finite steps during compression from initial volume, V_i to final volume, V_f
(c) Both (a) and (b)
(d) Cannot say

Sol: It is an irreversible compression. If the pressure is not constant at every stage of compression, but changes in number of finite steps, work done on the gas will be summed over all the steps and will be equal to $-\sum P\Delta V$.

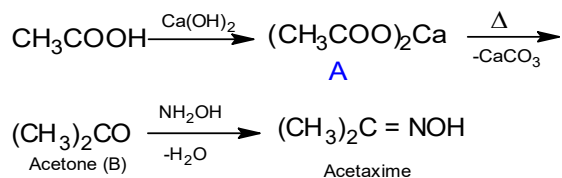
Ans: (c)

14. The end product 'C' in the following sequence of chemical reactions is



- (a) acetaldehyde oxime (b) formaldehyde oxime
(c) methyl nitrate (d) acetoxime

Sol:



Ans: (d)

15. Why is HCl not used to make the medium acidic in oxidation reactions of KMnO_4 in acidic medium?

- (a) Both HCl and KMnO_4 act as oxidising agents.
(b) KMnO_4 oxidises HCl into Cl_2 which is also a oxidising agent.
(c) KMnO_4 is a weaker oxidising agent than HCl.
(d) KMnO_4 acts as a reducing agent in the presence of HCl.

Sol: KMnO_4 oxidises HCl into Cl_2 which is also a oxidising agent.

Ans: (b)

16. Increasing pK_b values of *o*-, *m*- and *p*-toluidines is

- (a) $p < m < o$ (b) $o < m < p$ (c) $p < o < m$ (d) $m < o < p$

Sol: The basicity of *o*-, *m*- and *p*-toluidines decreases in the order $p > m > o$. In other words, their pK_b values will increase in the reverse order, i.e., option (a) is correct.

Ans: (a)

17. When a catalyst increase the rate of a chemical reaction, the rate constant?

- (a) remains constant
(b) increases
(c) decreases
(d) May increase (or) decrease depending on the order of reaction

Sol: $r = k [\text{Reactant}]$

at the same Concentration rate $\uparrow \rightarrow k \uparrow$

(or)

$$K = Ae^{-E_a/RT}$$

Presence of catalyst E_a decreases, Hence k increases

Ans: (b)

18. Graphite is a good conductor of electricity due to the presence of

- (a) lone pair of electrons (b) free valence electrons
(c) cations (d) anions

Sol: In graphite each C atom is sp^2 hybridised. One electron of each C atom remains free and makes graphite a good conductor of electricity.

Ans: (b)

19. What is the free energy change for the half reaction $Li^+ + e^- \rightarrow Li$?

Given $E^\circ_{Li^+/Li} = -3.0V$, $F = 96500 \text{ C mol}^{-1}$ and $T = 298K$.

- (a) $289.5 \text{ kJ mol}^{-1}$ (b) -298 kJ mol^{-1} (c) $32.166 \text{ kJ mol}^{-1}$ (d) $-289500 \text{ kJ mol}^{-1}$

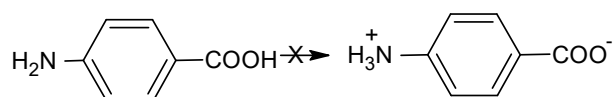
Sol: $\Delta G^\circ = -nFE^\circ = -1 \times 96500 \times (-3.0) \text{ J} = 289.5 \text{ kJ}$

Ans: (a)

20. Which of the following does not exist as a zwitterion?

- (a) Glycine (b) Alanine
(c) Sulphanilic acid (d) *p*-Aminobenzoic acid

Sol:



Ans: (d)

21. The half life of a first order reaction is 30 min and the initial concentration of the reactant is 0.1 M. If the initial concentration of reactant is doubled, then the half life of the reaction will be

- (a) 1800 s (b) 60 min (c) 15 min (d) 900 s

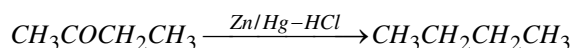
Sol: Half life of a first order reaction is independent of the initial concentration

Ans: (a)

22. Which of the following is reduced with zinc and hydrochloric acid to give the corresponding hydrocarbon?

- (a) ethyl acetate (b) acetic acid (c) acetamide (d) butan-2-one

Sol: Only ketones are reduced by Zn/HCl (Clemmenses reduction) to the corresponding hydrocarbons



Ans: (d)

23. 6.02×10^{20} molecules of urea are present in 100ml of its solution. The Concentration of the solution is

- (a) 0.02M (b) 0.01M (c) 0.001M (d) 0.1M

Sol: $\therefore$ No. of moles of Urea = $\frac{6.02 \times 10^{20}}{6.02 \times 10^{23}} = 10^{-3} \text{ mol}$

$\therefore$ Molar Concentration of urea.

$$= \frac{10^{-3}}{100} \times 1000$$

$$= 10^2 \text{ m}$$

$$= 0.01 \text{ M}$$

Ans: (b)

24. Pure benzene freezes at 5.3°C . A solution of 0.223 g of phenylacetic acid ($\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$) in 4.4 g of benzene ($K_f = 5.12 \text{ K kg mol}^{-1}$) freezes at 4.47°C . From this observation, one can conclude that

- (a) phenylacetic acid exists as such in benzene
- (b) phenylacetic acid undergoes partial ionization in benzene
- (c) phenylacetic acid undergoes complete ionisation in benzene
- (d) phenylacetic acid dimerises in benzene.

Sol: $\Delta T_f = T_f^\circ - T_f = 5.3 - 4.47 = 0.83$

Molality of solvent (m) = $\frac{0.223}{136} \times \frac{1000}{4.4} = 0.373$

(molecular weight of phenylacetic acid = 136)

$K_f = 5.12 \text{ K kg mol}^{-1}$

$\Delta T_f = i \cdot K_f \cdot m$

$i = \frac{\Delta T_f}{K_f \cdot m} = \frac{0.83}{5.12 \times 0.373} = 0.434$

$i < 1$ which means phenylacetic acid undergoes dimerization in benzene

Ans: (d)

25. The time taken for 10% completion of a first order reaction is 20 mins. Then for 19% completion, the reacting will take,

- (a) 40 mins
- (b) 60 mins
- (c) 30 mins
- (d) 50 mins

Sol: $t = \frac{2.303}{k} \log \frac{a}{a-x}$

$\therefore 20 = \frac{2.303}{k} \log \frac{a}{a-0.10a}$

$20 = \frac{2.303}{k} \log \frac{100}{90} \rightarrow (1)$

In Second case $t = \frac{2.303}{k} \log \frac{a}{a-0.19a}$

$t = \frac{2.303}{k} \log \frac{100}{81} \rightarrow (2)$

$\therefore (2)/(1) \Rightarrow \frac{t}{20} = 2$

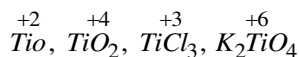
$\therefore t = 40 \text{ mins}$

Ans: (a)

26. The titanium (atomic number 22) Compound that does not exist is

- (a) TiO (b) TiO_2 (c) TiCl_3 (d) K_2TiO_4

Sol:



$\text{Ti} = +6$ oxidation state not possible

Ans: (d)

27. The electrode potential for $\text{Cu}^{+2}_{(aq)} + e^- \rightarrow \text{Cu}^+_{(aq)}$ and $\text{Cu}^+_{(aq)} + e^- \rightarrow \text{Cu}_{(s)}$ are +0.15V and +0.50V

respectively. The value of $E^0_{\text{Cu}^{+2}/\text{Cu}}$ will be

- (a) 0.150V (b) 0.500V (c) 0.325V (d) 0.650V



$$\therefore \Delta G_3 = \Delta G_1 + \Delta G_2$$

$$\therefore E^0_{\text{Cu}^{+2}/\text{Cu}} = \frac{0.65}{2} = 0.325\text{V}$$

Ans: (c)

28. On heating with soda-lime, salicylic acid gives

- (a) Phenol (b) Benzoic acid (c) Sodium salicylate (d) Benzene

Sol: Phenol

Ans: (a)

29. Red hot carbon will remove oxygen from the oxide AO and BO but not from MO , while B will remove oxygen from AO . The activity of metals A, B and M in decreasing order is

- (a) $A > B > M$ (b) $B > A > M$ (c) $M > B > A$ (d) $M > A > B$

Sol: Carbon removes oxygen from AO and BO , C is better reducing agent than A and B ; B removes oxygen from AO , B is better reducing agent than A . Better reducing means more active. Hence the correct order is $M < B < A$.

Ans: (c)

30. which is the strongest base among the following?

- (a) $\text{La}(\text{OH})_3$ (b) $\text{Lu}(\text{OH})_3$ (c) $\text{Ce}(\text{OH})_3$ (d) $\text{Yb}(\text{OH})_3$

Sol: $\text{La}(\text{OH})_3$ [$\text{La} - \text{Lu}$ basic Strength decreases]

Ans: (a)

31. Actinoid contraction is greater from element to element than lanthanoid contraction, because
- 5f -electrons themselves provide poor shielding from element to element
 - 4f -electrons provide poor shielding as compared to 5f -electrons
 - both (a) and (b)
 - neither (a) nor (b)

Sol: 5f -electrons themselves provide poor shielding from element to element

Ans: (a)

32. 60 g of urea (mol. wt. 60) was dissolved in 9.9 moles of water. If the vapour pressure of pure water is p°_1 , the vapour pressure of solution is
- 0.10 p°_1
 - 1.10 p°_1
 - 0.90 p°_1
 - 0.99 p°_1

Sol: $\frac{p^\circ_1 - p}{p^\circ_1} = \frac{\text{Moles of urea}}{\text{Moles of water}}$

$$\frac{p^\circ_1 - p}{p^\circ_1} = \frac{1}{9.9} \Rightarrow p = 0.89 p^\circ_1 = 0.9 p^\circ_1$$

$$\left[\because n_{\text{urea}} = \frac{W}{M} = \frac{60}{60} = 1 \right]$$

Ans: (c)

33. Acetone on distillation with conc. H_2SO_4 forms
- phorone
 - acrolein
 - mesitylene
 - mesityl oxide

Sol: Mesitylene, i.e., 1, 3, 5-trimethylbenzene is formed

Ans: (c)

34. The half-life period of a radioactive element is 20 days. What will be the remaining mass of 100g of it after 60 days?
- 25g
 - 50g
 - 12.5g
 - 20g

Sol:

$$60 \text{ days} = 3 \text{ half-lives } (n = 3)$$

$$[A] = \frac{[A]_0}{2^n} = \frac{100}{2^3} = 12.5 \text{g}$$

Ans: (c)

35. On the basis of information available from the reaction $\frac{4}{3} Al + \frac{3}{2} O_2 \rightarrow Al_2O_3, \Delta G = -827 \text{ kJ mol}^{-1}$ of O_2 .

'The minimum emf required to carry out an electrolysis of Al_2O_3 is ($F = 96500 \text{ C mol}^{-1}$)

- 2.14 V
- 4.28 V
- 6.42 V
- 8.56 V

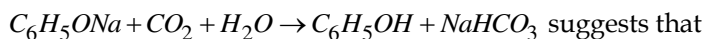
Sol: $2Al \rightarrow 2Al^{3+} + 6e^-$; $4/3 Al \rightarrow 4Al^{3+} + 4e^-$

For $\frac{4}{3}$ mol of Al to change into Al^{3+} ions, $n = 4$. Hence, $\Delta G = -nFE$ gives $E = \frac{\Delta G}{-nF} = \frac{-827000}{-4 \times 96500} = 2.14$

volt

Ans: (a)

36. The reaction,

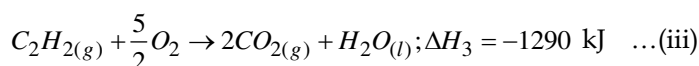
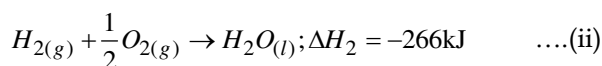


- (a) phenol is a stronger acid than carbonic acid
- (b) carbonic acid is a stronger acid than phenol
- (c) water is a stronger acid than phenol
- (d) phenol is stronger acid than water.

Sol: Since a stronger acid displaces a weaker acid from its salts, therefore, carbonic acid ($CO_2 + H_2O = H_2CO_3$) is a stronger acid than phenol.

Ans: (b)

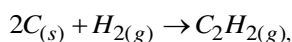
37. Given: $2C_{(s)} + 2O_{2(g)} \rightarrow 2CO_{2(g)}; \Delta H_1 = -767 \text{ kJ} \dots(i)$



Heat of formation of acetylene is

- (a) 1802 kJ
- (b) - 1802 kJ
- (c) 257 kJ
- (d) 237 kJ

Sol: The required reaction is



Applying the operation (i) + (ii) - (iii), we get By Hess's law heat summation;

$$\Delta H = \Delta H_1 + \Delta H_2 - \Delta H_3 = (-767) + (-266) - (-1290) = 257 \text{ kJ}.$$

Ans: (c)

38. The ease of dehydrohalogenation of alkyl halides follows the order :

- (a) $1^\circ > 2^\circ > 3^\circ$
- (b) $2^\circ > 3^\circ > 1^\circ$
- (c) $3^\circ > 1^\circ > 2^\circ$
- (d) $3^\circ > 2^\circ > 1^\circ$

Sol: The ease of dehydrohalogenation of alkyl halides is :

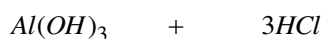
$$3^\circ > 2^\circ > 1^\circ$$

Ans: (d)

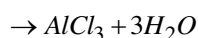
39. Gastric juice contains about 3.0 g of HCl per litre. If a person produces about 2.5 litre of gastric juice per day, number of antacid tablets each containing 400 mg of $Al(OH)_3$ needed to neutralize all the HCl produced per day will be

- (a) 10
- (b) 18
- (c) 14
- (d) 8

Sol: HCl is neutralized by $Al(OH)_3$ according to the equation,



$$27 + 3 \times 17 = 78 \text{ g} \quad 3 \times 36.5 = 109.5 \text{ g}$$



HCl produced from 2.5 litre of gastric juice

$$= 3.0 \times 2.5 = 7.5 \text{ g}$$

109.5 g HCl is neutralized by $Al(OH)_3 = 78 \text{ g}$

$\therefore 7.5g$ HCl will be neutralized by $Al(OH)_3$

$$= \frac{78}{109.5} \times 7.5g = 5.342g = 5342 \text{ mg}$$

$$\therefore \text{Number of tablets required} = \frac{5342}{400} = 13.35,$$

i.e., 14 tablets.

Ans: (c)

40. The osmotic pressure of 0.1M aqueous solution of NaCl is ----- Osmotic pressure of 0.1 M aqueous solution of glucose

(a) Equal to (b) less than (c) half of (d) nearly double

Sol: $NaCl \rightarrow Na^+ + Cl^-$

Hence Concentration of particle in 0.1M NaCl solution will be double, and Osmotic pressure will also be double

Ans: (d)

41. Solubility of a gas in a liquid increases with

(a) increase of pressure and increase of temperature
(b) decrease of pressure and increase of temperature
(c) increase of pressure and decrease of temperature
(d) decrease of pressure and decrease of temperature

Sol: increase of pressure and decrease of temperature

Ans: (c)

42. Consider the following:

1. EDTA - Hexadentate legand
2. ethylenediamine - Bidentate ligand
3. diethylenetriamine - Tridentate ligand
4. triethylenetetraamine - Tridentate ligand

Which is / are not correctly paried?

(a) 1 and 2 (b) 3 and 4 (c) only 4 (d) 1, 2 and 3

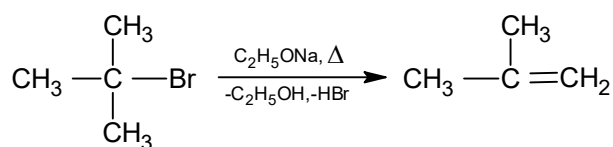
Sol: It is a tetradentate ligand

Ans: (c)

43. The major product obtained when tert-butyl bromide is heated with sodium ethoxide is

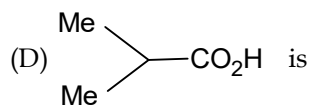
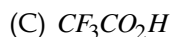
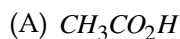
(a) 2-methyl-1-propene (b) ethene
(c) tert-butyl methyl ether (d) diethyl ether

Sol: 3° Alkyl halides on treatment with an alkoxide ion tend to undergo elimination.



Ans: (a)

44. The correct order of increasing acid strength of the compounds

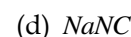
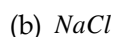


Sol: CF_3 is a strong electron - withdrawing group but MeO is a weak electron - withdrawing group, therefore, $CF_3COOH(C)$ is a stronger acid than $MeOCH_2COOH(B)$. Further, $(Me)_2CH$ is a stronger electron - donating group than CH_3 , therefore, $CH_3COOH(A)$ is a stronger acid than $(Me)_2CHCO_2H(D)$. Thus, the overall acid character increases in the order:

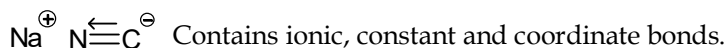
$D < A < B < C$, i.e., Option (c) is correct

Ans (c)

45. Which one of the following contains ionic, covalent and coordinate bonds?

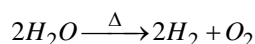


Sol:



Ans: (c)

46. In the following decomposition reaction

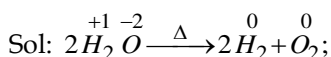


(a) hydrogen is oxidised

(b) oxygen is reduced

(c) hydrogen is reduced

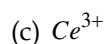
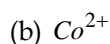
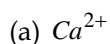
(d) oxygen remains unaffected



here hydrogen is reduced

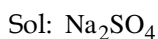
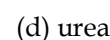
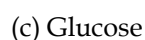
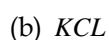
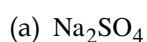
Ans: (c)

47. Which of the following metal ions has highest tendency to form complex?



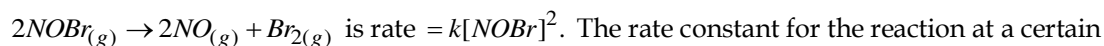
Ans: (b)

48. Which of the following 0.1M aqueous solution is likely to have the highest boiling point?



Ans: (a)

49. The rate law for the reaction,



temperature is $1.62\text{M}^{-1}\text{s}^{-1}$. If the concentration of NOBr at a certain time is $2.00 \times 10^{-3}\text{M}$, what will be the rate of reaction?

- (a) $4.40 \times 10^{-6}\text{M s}^{-1}$ (b) $4.50 \times 10^{-6}\text{M s}^{-1}$ (c) $6.44 \times 10^{-6}\text{M s}^{-1}$ (d) $6.48 \times 10^{-6}\text{M s}^{-1}$

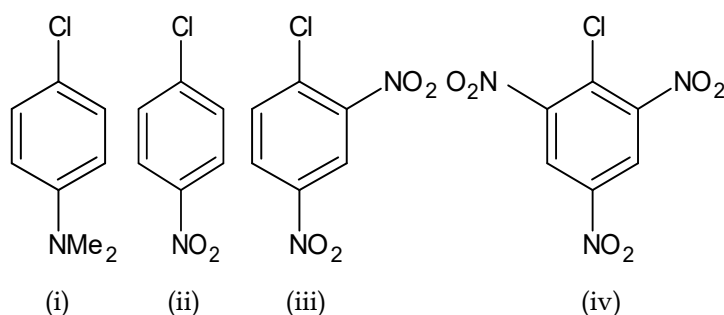
Sol: Rate = $k[\text{NOBr}]^2$

$$k = 1.62\text{M}^{-1}\text{s}^{-1}, [\text{NOBr}] = 2.00 \times 10^{-3}\text{M}$$

$$\text{Hence, rate} = 1.62(\text{M}^{-1}\text{s}^{-1}) \times (2 \times 10^{-3})^2(\text{M}^2) = 6.48 \times 10^{-6}\text{M s}^{-1}$$

Ans: (d)

50. Order of reactivity towards nucleophilic substitution reaction of the compounds



- (a) $i > ii > iii > iv$ (b) $ii > i > iii > iv$ (c) $iv > iii > ii > i$ (d) $iii > iv > ii > i$

Sol: Electron - withdrawing groups increase and electron - donating groups decrease the reactivity towards nucleophilic substitution. Thus, option (c). $iv > iii > ii > i$ is correct.

Ans: (c)

51. Which one of the following is wrongly matched?

- (a) $[\text{Cu}(\text{NH}_3)_4]^{+2} \rightarrow$ Square planar?
 (b) $[\text{Ni}(\text{CO})_4] \rightarrow$ neutral ligand
 (c) $[\text{Fe}(\text{CN})_6]^{-3} \rightarrow sp^3d^2$
 (d) $[\text{Co}(\text{en})_3]^{+3} \rightarrow$ follows EAN rule

Sol: $[\text{Fe}(\text{CN})_6]^{-3} \rightarrow$ Inner orbital octahedral Complex, with the hybridization d^2sp^3

Ans: (c)

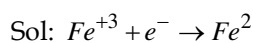
52. In Lassaigne's test, the sodium extract of an organic compound containing both nitrogen and sulphur on treatment with FeCl_3 , produces a blood red colouration due to the formation of

- (a) $\text{Fe}(\text{CNS})_2$ (b) NaCNS (c) NH_4CNS (d) $\text{Fe}(\text{SCN})_3$

Sol: Ferric thocyanate or sulphocyanide, $\text{Fe}(\text{SCN})_3$.

Ans: (d)

53. Calculate the reduction potential of a half cell consisting of platinum electrode immersed in $2.0\text{ M } Fe^{+2}$ and $0.02\text{ M } Fe^{+3}$, given $E^0_{Fe^{+3}/Fe^{+2}} = 0.771\text{ V}$
- (a) 0.653 V (b) 0.889 V (c) 0.683 V (d) 2.771 V



$$E_{red} = E^0_{Fe^{+3}/Fe^{+2}} - \frac{0.0591}{1} \log \frac{Fe^{+2}}{Fe^{+3}}$$

$$= 0.771 - 0.0591 \times \log \frac{2.0}{0.02}$$

$$= 0.771 - 0.0591 \times \log 100$$

$$= 0.653\text{ V}$$

Ans: (a)

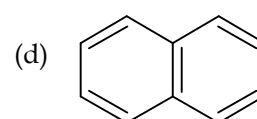
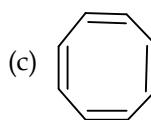
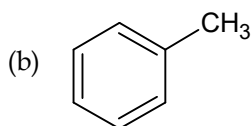
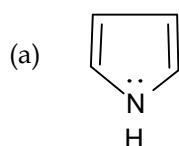
54. Mercury is a liquid metal because

- (a) it has a completely filled s - orbital
 (b) it has a small atomic size
 (c) it has a completely filled d -orbital that prevents $d-d$ overlapping of orbitals
 (d) it has a completely filled d -orbital that causes $d-d$ overlapping

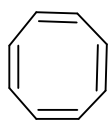
Sol: The electronic configuration of mercury (80) is, $[Xe]4f^{14}, 5d^{10}, 6s^2$. Its d -subshell is completely filled, thus it prevents the overlapping of d -orbitals ($d-d$ overlapping). Hence, it is liquid metal at room temperature.

Ans: (c)

55. Which is a non-aromatic compound?



Sol:



is non-aromatic

Ans: (c)

56. The pH of the solution prepared by mixing 2.0 ml of HCl solution of $pH\ 3.0$ and 3.0 ml of $NaOH$ solution of $pH\ 10.0$ is

- (a) 2.5 (b) 3.5 (c) 5.5 (d) 6.5

Sol: 2 ml of HCl with $pH\ 3$, i.e., $[H^+] = 10^{-3}\text{ M}$ contains 2×10^{-3} millimole or millieq of HCl . 3 ml of $NaOH$ with $pH\ 10$, i.e. $[H^+] = 10^{-10}\text{ M}$ or $[OH^-] = 10^{-4}\text{ M}$ contains 3×10^{-4} millimole or millieq of $NaOH$.

$$\text{HCl left unneutralized} = 2 \times 10^{-3} - 3 \times 10^{-4}$$

$$(20 \times 10^{-4} - 3 \times 10^{-4}) = 17 \times 10^{-4}$$

$$\text{Volume of solution} = 2 + 3 \text{ ml} = 5 \text{ ml}$$

$$\therefore [H^+] = \frac{17 \times 10^{-4}}{5} = 3.4 \times 10^{-4} \text{ M}$$

$$pH = -\log(3.4 \times 10^{-4})$$

$$= 4 - 0.53 = 3.47 \approx 3.5$$

Ans: (b)

57. For the reaction $C(s) + CO_2(g) \rightarrow 2CO(g)$, $K_p = 63$ atm at 100 K. If at equilibrium,

$p_{CO} = 10p_{CO_2}$, then the total pressure of the gases at equilibrium is

- (a) 6.3 atm (b) 6.93 atm (c) 0.63 atm (d) 0.693 atm

Sol: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

$$K_p = \frac{p_{CO}^2}{p_{CO_2}} \text{ or } 63 = \frac{(10p_{CO_2})^2}{p_{CO_2}}$$

$$\text{or } p_{CO_2} = \frac{63}{100} = 0.63 \text{ atm}$$

$$p_{CO} = 10p_{CO_2} = 10 \times 0.63 \text{ atm} = 6.3 \text{ atm}$$

$$\therefore p_{\text{total}} = p_{CO_2} + p_{CO} = 0.63 + 6.3 = 6.93 \text{ atm}$$

Ans: (b)

58. In chromite ore, the oxidation number of iron and chromium are respectively

- (a) +3, +2 (b) +3, +6 (c) +2, +6 (d) +2, +3

Sol: Chromite ore is $FeCr_2O_4$

Ox state of $Fe = +2$

$Cr = +3$

Ans: (d)

59. Which of the following can possibly be used as analgesic without causing addiction and modification?

- (a) Morphine (b) N-Acetyl-*para*-aminophenol
(c) Diazepam (d) Tetrahydrocatecol

Sol: N-Acetyl-*para*-aminophenol is the chemical name of paracetamol

Ans: (b)

60. Which of the following Carbonyls will have the strongest $C-O$ bond?

- (a) $[Mn(CO)_6]^+$ (b) $Cr(CO)_6$ (c) $V(CO)_6^-$ (d) $Fe(CO)_5$

Sol: As the charge on the Central atom increases, the metal shows lessor tendency to donate its electron density to the antibonding π orbital of CO ligand. This causes strengthening of the $C-O$ bond.

Ans: (a)

Mathematics

Multiple Choice Questions with one correct answer. A correct answer carries 1 mark. No negative mark. **60 x 1 = 60**

61. The value of $\int_0^2 |1-x| dx$ is

Options:

(a) 2

(b) $\frac{3}{2}$

(c) 0

(d) 1

$$\begin{aligned} \text{Sol: } I &= \int_0^1 |1-x| dx + \int_1^2 |1-x| dx = \int_0^1 (1-x) dx + \int_1^2 (x-1) dx \\ &= x - \frac{x^2}{2} \Big|_0^1 + \frac{x^2}{2} - x \Big|_1^2 = 1 - \frac{1}{2} + \left[2 - 2 - \left(\frac{1}{2} - 1 \right) \right] = \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

Ans: (d)

62. The area enclosed by the curve $x^2 = 4ay$, the x - axis, and the ordinates $x = 0, x = 4a$, in square units, is

Options:

(a) $16a^2$

(b) $\frac{16a^2}{3}$

(c) $3a^2$

(d) $48a^2$

$$\text{Sol: } \int_0^{4a} \frac{x^2}{4a} dx = \frac{1}{4a} \left[\frac{x^3}{3} \right]_0^{4a} = \frac{1}{4a} \cdot \frac{1}{3} [64a^3] = \frac{16a^2}{3}$$

Ans: (b)

63. The area of the region bounded by $y = 2x - x^2$ and the x - axis is

Options:

(a) $\frac{7}{3}$ sq. units

(b) $\frac{2}{3}$ sq. units

(c) $\frac{8}{3}$ sq. units

(d) $\frac{4}{3}$ sq. units

Sol: $y = 2x - x^2, y = 0 \Rightarrow 2x - x^2 = 0 \Rightarrow x = 0, x = 2$

Required area

$$= \int_0^2 y dx = \int_0^2 (2x - x^2) dx = x^2 \Big|_0^2 - \frac{x^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3} \text{ sq.units}$$

Ans: (d)

64. The solution of the differential equation $y dx + (x + x^2 y) dy = 0$ is

Options:

(a) $\frac{1}{xy} + \ln y = c$

(b) $-\frac{1}{xy} + \ln y = c$

(c) $\frac{-1}{xy} = c$

(d) $\ln y = cx$

Sol: Given equation is $\frac{dx}{dy} + \frac{x + x^2 y}{y} = 0$

Dividing by $-x^2$

$$\Rightarrow -\frac{1}{x^2} \cdot \frac{dx}{dy} + \frac{(x + x^2 y)}{-x^2 y} = 0 \Rightarrow -\frac{1}{x^2} \cdot \frac{dx}{dy} + \frac{1}{y} \left(-\frac{1}{x} \right) = 1$$

Putting $\frac{1}{x} = t \Rightarrow \frac{-1}{x^2} \frac{dx}{dy} = \frac{dt}{dy} \Rightarrow \frac{dt}{dy} + \frac{1}{y} (-t) = 1 \Rightarrow \frac{dt}{dy} - \frac{1}{y} \cdot t = 1$

This is linear in t

Integrating factor $= e^{\int -\frac{1}{y} dy} = e^{-\log_e y} = \frac{1}{y}$

Solution is $t \cdot \frac{1}{y} = \int 1 \times \frac{1}{y} \cdot dy + c$

$$\Rightarrow \frac{t}{y} = \ln y + c \quad \text{But } t = \frac{1}{x}$$

$$\therefore \frac{1}{xy} = \ln y + c$$

Ans: (b)

65. The order and degree of the differential equation $x \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 + y^2 = 0$ are respectively

Options:

- (a) 2 and 1
- (b) 1 and 2
- (c) 2 and 2
- (d) 1 and 1

Sol: Consider $x \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 + y^2 = 0$

Order of the differential equation is 2 and degree of the differential equation is 1.

Ans: (a)

66. Solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}$ satisfying the condition $y(1) = 1$ is

Options:

- (a) $y = \log x + x$
- (b) $y = x \log x + x^2$
- (c) $y = x e^{x-1}$
- (d) $y = x \log x + x$

Sol: $\frac{dy}{dx} = \frac{x+y}{x}$ This is a homogenous differential equation.

Putting $y = vx$

$$v + x \frac{dv}{dx} = \frac{x+vx}{x} \quad x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x}$$

$$\int dv = \int \frac{1}{x} dx$$

$$v = \log_e x + c$$

$$\frac{y}{x} = \log x + c$$

$$\text{When } x=1 \Rightarrow y=1 \therefore c=1 \quad \therefore y = x \log x + x$$

Ans: (d)

67. The shortest distance between the lines $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$ is

Options:

- (a) 6
- (b) $\sqrt{5}$
- (c) $\frac{6}{\sqrt{5}}$
- (d) $6\sqrt{5}$

Sol: $\vec{a}_2 - \vec{a}_1 = (-3\hat{i} + 0\hat{j} - 2\hat{k})$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix} = \hat{i}(-10+12) - \hat{j}(-5+6) + \hat{k}(4-4)$$

$$= 2\hat{i} - \hat{j}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{4+1} = \sqrt{5}$$

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = -6 + 0 + 0$$

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{|-6|}{\sqrt{5}} = \frac{6}{\sqrt{5}}$$

Ans: (c)

68. Vector equation of the line $6x - 2 = 3y + 1 = 2z - 2$ is $\vec{r} =$

Options:

(a) $\hat{i} - \hat{j} + 3\hat{k} + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$

(b) $\hat{i} + 2\hat{j} + 3\hat{k} + \lambda\left(\frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} + \hat{k}\right)$

(c) $\frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} + \hat{k} + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$

(d) $-2\hat{i} + \hat{j} - 2\hat{k} + \lambda(6\hat{i} + 3\hat{j} + 2\hat{k})$

Sol: The equations of the line can be written as

$$\frac{x - \frac{1}{3}}{\frac{1}{6}} = \frac{y + \frac{1}{3}}{\frac{1}{3}} = \frac{z - 1}{\frac{1}{2}}$$

$\therefore$ The line passes through $\left(\frac{1}{3}, -\frac{1}{3}, 1\right)$ and D.C.'s are

proportional to 1, 2, 3.

$\therefore$ The required equation is

$$\vec{r} = \frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} + \hat{k} + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$$

Ans: (c)

69. Equations of a line through $(1, 2, -2)$ and parallel to the line whose equations are $\frac{x-3}{2} = \frac{y-1}{7} = \frac{z+3}{-2}$

Options:

(a) $\frac{x-1}{2} = \frac{y-2}{7} = \frac{z+2}{-2}$

(b) $\frac{x-2}{3} = \frac{y-7}{1} = \frac{z+2}{-3}$

(c) $\frac{x-2}{2} = \frac{y-3}{-1} = \frac{z+3}{1}$

(d) $\frac{x-3}{2} = \frac{y+1}{-1} = \frac{z=2}{1}$

Sol: The line passes through $(1, 2, -2)$ and its direction cosines are proportional to $2, 7, -2$.

$\therefore$ Its equation are $\frac{x-1}{2} = \frac{y-2}{7} = \frac{z+2}{-2}$

Ans: (a)

70. The minimum value of $z = 5x - 4y$ subject to constraints $x + y \leq 10, y \leq 4; x, y > 0$ will be at the point

Options:

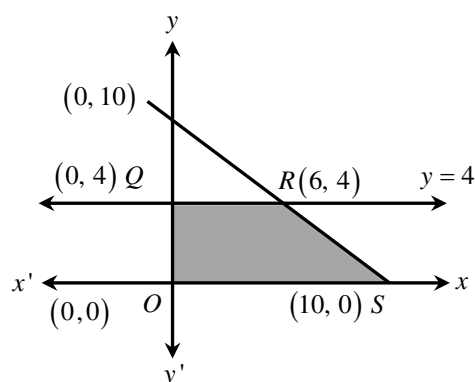
(a) $(10, 4)$

(b) $(-10, 4)$

(c) $(6, 4)$

(d) $(0, 4)$

Sol:



$\therefore$ $QRSO$ is the feasible region and $z = 5x - 4y$

At $Q(0, 4)$ $z = -16$

At $R(6, 4)$ $z = 14$

At $O(0, 0)$ $z = 0$

At $S(10, 0)$ $z = 50$

$\therefore$ Minimum value is -16 which is attained at point $(0, 4)$

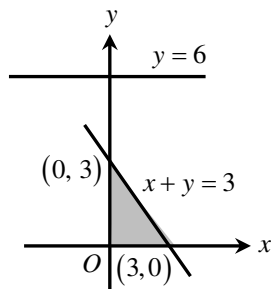
Ans: (d)

71. The region represented by the system of equations $x, y \geq 0, y \leq 6, x + y \leq 3$ is

Options:

- (a) unbounded in first quadrant
- (b) unbounded in first and second quadrants
- (c) bounded in first quadrant
- (d) bounded in 2nd quadrant

Sol: The given region is bounded in first quadrant.



Ans: (c)

72. If A is a square matrix such that $A^2 = A$ then $(I + A)^3 - 7A$ is equal to

Options:

- (a) A
- (b) $I - A$
- (c) I
- (d) $3A$

$$\begin{aligned} \text{Sol: } (I + A)^2 &= (I + A)(I + A) = I^2 + IA + AI + A^2 \\ &= I + A + A + A \\ &= I + 3A \quad \left(\because A^2 = A \text{ given} \right) \end{aligned}$$

$$\begin{aligned} (I + A)^3 &= (I + A)^2 (I + A) = (I + 3A)(I + A) \\ &= I^2 + IA + 3AI + 3A^2 = I + A + 3A + 3A = I + 7A \\ \therefore (I + A)^3 - 7A &= I \end{aligned}$$

Ans: (c)

73. If $A+B=\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $A-2B=\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ then A is

Options:

(a) $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$

(c) $\begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

(d) $\begin{bmatrix} \frac{1}{3} & 0 \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$

Sol: Multiplying the first matrix by 2 and adding the same to the second matrix, we get

$$2A+2B=\begin{bmatrix} 2 & 0 \\ 2 & 2 \end{bmatrix}, A-2B=\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \Rightarrow 3A=\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$\Rightarrow A=\begin{bmatrix} \frac{1}{3} & 0 \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Ans: (d)

74. If $A=\begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$, then A^4 in terms of A is

Options:

(a) $14A-15I$

(b) $15A-14I$

(c) $5A-12I$

(d) $4A-15I$

Sol: Consider the determinant $\Delta=\begin{vmatrix} -1 & 3 \\ -2 & 4 \end{vmatrix}$

Its characteristic equation is

$$\begin{vmatrix} -1-\lambda & 3 \\ -2 & 4-\lambda \end{vmatrix}=0 \Rightarrow -(1+\lambda)(4-\lambda)+6=0 \Rightarrow \lambda^2-3\lambda+2=0$$

Replacing λ by A , we get

$$A^2-3A+2I=0 \Rightarrow A^2=3A-2I$$

$$A^4=(3A-2I)^2$$

$$=9A^2-12A+4I=9(3A-2I)-12A+4I \quad [\text{using (1)}]$$

$$= 27A - 18I - 12A + 4I = 15A - 14I.$$

Ans: (b)

75. The value of $\begin{vmatrix} 43 & 44 & 45 \\ 44 & 45 & 46 \\ 45 & 46 & 46 \end{vmatrix}$ is

Options:

(a) 0

(b) -1

(c) 1

(d) 2

Sol: $\begin{vmatrix} 43 & 44 & 45 \\ 44 & 45 & 46 \\ 45 & 46 & 46 \end{vmatrix}$ On defining $R'_1 = R_1 - R_2$ and

$$R'_2 = R_2 - R_3,$$

we get $\begin{vmatrix} -1 & -1 & -1 \\ -1 & -1 & 0 \\ 45 & 46 & 46 \end{vmatrix}$

On defining

$$C'_1 = C_1 - C_2 \text{ and } C'_2 = C_2 - C_3 \begin{vmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 46 \end{vmatrix}$$

$$= (-1)(-1) = 1.$$

Ans: (c)

76. Which one of the following is one – one?

Options:

(a) $f(x) = x^2$

(b) $f(x) = \frac{x^2 - 1}{2}$

(c) $f(x) = 2x + 5$

(d) $f(x) = \cos x. (0 \leq x \leq 2\pi)$

Sol: $f(x) = x^2$ is not one-one because $f(x_1) = f(x_2)$ does not $\Rightarrow x_1 = x_2$

Similarly, $f(x) = \frac{x^2 - 1}{2}$ is also not one-one.

$$\text{If } f(x) = 2x + 5, f(x_1) = f(x_2) \Rightarrow 2x_1 + 5 = 2x_2 + 5 \Rightarrow x_1 = x_2.$$

Thus, $f(x) = 2x + 5$ is one-one.

If $f(x) = \cos x (0 \leq x \leq 2\pi)$ is not one-one.

Ans: (c)

77. Let $f(x) = \begin{cases} -2\sin x & \text{if } x \leq -\frac{\pi}{2} \\ a \sin x + b & \text{if } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x & \text{if } x \geq \frac{\pi}{2} \end{cases}$. The values of a and b so that $f(x)$ is continuous are

Options:

- (a) $a=1, b=1$
- (b) $a=1, b=-1$
- (c) $a=-1, b=1$
- (d) $a=-1, b=-1$

Sol: LHL at $x = -\frac{\pi}{2}$ is 2 and RHL at $x = -\frac{\pi}{2}$ is $-a+b$

$f(x)$ is continuous at $x = -\frac{\pi}{2}$ if $-a+b=2$... (1)

LHL at $x = \frac{\pi}{2}$ is $a+b$ and RHL at $x = \frac{\pi}{2}$ is 0

$f(x)$ is continuous at $x = \frac{\pi}{2}$ if $a+b=0$... (2)

From (1) and (2).

$$a = -1, b = 1.$$

Ans: (c)

78. $f(x) = \sin x + \log_e x$ is continuous

Options:

- (a) $\forall x$
- (b) $\forall x > 0$
- (c) $\forall x < 0$
- (d) $\forall x \in (0, 1]$

Sol: $f(x) = \sin x + \log_e x$

$\log_e x$ is continuous at all the points $x > 0$.

$\therefore \log$ (of negative number) is not defined.

Ans: (b)

79. Let $[]$ denote the greatest integer function and $f(x) = [\tan^2 x]$ then

Options:

- (a) $\lim_{x \rightarrow 0} f(x)$ does not exist
- (b) $f(x)$ is continuous at $x = 0$
- (c) $f(x)$ is not differentiable at $x = 0$
- (d) $f'(0) = 1$

Sol: In the neighbourhood of $x = 0, 0 \leq \tan^2 x < 1$.

$$\therefore [\tan^2 x] = 0.$$

($\because$ integral part of any number between 0 and 1 is 0)

$$\therefore f(x) = 0, \forall x \text{ in the neighbourhood of } x = 0.$$

$\therefore f(x)$ is differentiable at $x = 0$, continuous at $x = 0$ and $f'(0) = 0$

Ans: (b)

80. The function $f(x) = |\sin x|$

Options:

- (a) differentiable at $x = 0$
- (b) not differentiable at $x = \frac{\pi}{2}$
- (c) not differentiable at $x = 0$
- (d) differentiable at $x = 0$ and $x = \frac{\pi}{2}$

Sol: For $x \in \left(-\frac{\pi}{2}, 0\right)$, $\sin x < 0$, $f(x) = -\sin x$,

$$f' = -\cos x, Lf'(0) = 0$$

For $x \in \left(0, \frac{\pi}{2}\right)$, $\sin x > 0$, $f(x) = \sin x$, $f'(x) = \cos x$,

$$Rf'(0) = 1$$

$$Lf'(0) \neq Rf'(0)$$

$\therefore f(x)$ is not differentiable at $x = 0$

Ans: (c)

81. If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ then $\frac{dy}{dx}$ is equal to

Options:

(a) $\frac{2}{1+x^2}$

(b) $\frac{3}{1+x^2}$

(c) $\frac{1}{1+x^2}$

(d) $\frac{-1}{1+x^2}$

Sol: $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = 2 \tan^{-1} x$

$$\therefore \frac{dy}{dx} = \frac{2 \times 1}{1+x^2} = \frac{2}{1+x^2}.$$

Ans: (a)

82. Derivative of $\sin^{-1}\left(2x\sqrt{1-x^2}\right)$ with respect to $\tan^{-1}\frac{x}{\sqrt{1-x^2}}$ is equal to; $\left(|x| < \frac{1}{\sqrt{2}}\right)$

Options:

(a) 1

(b) -1

(c) -2

(d) 2

Sol: By putting $x = \sin \theta$ in $y = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$ we get

$$\sin^{-1}(2 \sin \theta \cos \theta) = 2\theta$$

$$z = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) = \tan^{-1}\left(\frac{\sin \theta}{\cos \theta}\right) = \tan^{-1}(\tan \theta) = \theta$$

$$\therefore y = 2z, \quad \frac{dy}{dz} = 2.$$

Ans: (d)

83. The critical values of x of the function $f(x) = \sin^2 x$ in $[0, \pi)$ are

Options:

(a) $0, \pi$

(b) $0, \frac{\pi}{2}$

(c) $\frac{\pi}{3}, \frac{\pi}{2}$

(d) $\frac{\pi}{3}, \frac{2\pi}{3}$

Sol: $y = \sin^2 x \Rightarrow y_1 = 2 \sin x \cos x = 0$

$\sin x = 0 \Rightarrow x = 0, \cos x = 0 \Rightarrow x = \frac{\pi}{2}$

$\therefore$ The values are 0 or $\frac{\pi}{2}$.

Ans: (b)

84. The value of $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} e^{\sin x} \cdot \cos x \, dx$ is

Options:

(a) $e^{\sin x} + c$

(b) $e^{\cos x} + c$

(c) $\ln(\sin x - \cos x) + c$

(d) $e^{\sin x + \cos x} + c$

Sol: $I = \int \frac{\sin x + \cos x}{\sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cos x}} \cdot e^{\sin x} \cdot \cos x \, dx$

$= \int e^{\sin x} \cdot \cos x \, dx = e^{\sin x} + c$

Ans: (a)

85. The value of $\int \frac{dx}{\sqrt{(1-x^2) \sin^{-1} x}}$ is

Options:

(a) $(\sin^{-1} x)^2 + c$

(b) $\sqrt{\sin^{-1} x} + c$

(c) $2\sqrt{\sin^{-1} x} + c$

(d) $2(\sin^{-1} x)^2 + c$

Sol: $I = \int \frac{dx}{\sqrt{(1-x^2) \sin^{-1} x}}$

Taking $\sin^{-1} x = t, \frac{1}{\sqrt{1-x^2}} dx = dt$

$I = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + c = 2\sqrt{\sin^{-1} x} + c$

Ans: (c)

86. The value of $\int \frac{x + \sin x}{1 + \cos x} dx$ is

Options:

(a) $\tan \frac{x}{2} + c$

(b) $\cot \frac{x}{2} + c$

(c) $x \tan \frac{x}{2} + c$

(d) $\frac{1}{2} \cos \frac{x}{2} + c$

Sol: $I = \int \frac{x + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} dx = \frac{1}{2} \int x \sec^2 \frac{x}{2} dx + \int \tan \frac{x}{2} dx$

$\Rightarrow I = \int x \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) dx + \int \tan \frac{x}{2} dx \Rightarrow$

$x \tan \frac{x}{2} - \int 1 \cdot \tan \frac{x}{2} dx = x \tan \frac{x}{2} + c.$

Ans: (c)

87. The value of $\int \tan^2 2x dx$ is

Options:

(a) $\frac{\sec^2 2x}{2} - x + c$

(b) $\frac{\tan 2x}{2} + x + c$

(c) $\frac{\tan 2x}{2} - x + c$

(d) $\frac{1}{2} \cot 2x - x + c$

Sol: $\int \tan^2 2x dx = \int (\sec^2 2x - 1) dx = \frac{\tan 2x}{2} - x + c$

Ans: (c)

88. The value of $\int_{-1/2}^{1/2} \cos x \ln \left(\frac{1-x}{1+x} \right) dx$ is

Options:

(a) 1

(b) $2\sqrt{e}$

(c) 0

(d) $\sqrt{e}$

Sol: $\cos x$ is an even function and $\ln\left(\frac{1-x}{1+x}\right)$ is an odd function

$\therefore \cos x \cdot \ln\left(\frac{1-x}{1+x}\right)$ is an odd function. Thus, $I = 0$.

Ans: (c)

89. The value of $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx$ is

Options:

(a) $\frac{\pi(a+b)}{4}$

(b) $\frac{\pi(a-b)}{4}$

(c) $\frac{\pi(a+b)}{2}$

(d) $\frac{\pi(a-b)}{2}$

Sol: Let $I = \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx$

$$I = \int_0^{\pi/2} \frac{a \cos x + b \sin x}{\cos x + \sin x} dx$$

(1) + (2) gives

$$2I = \int_0^{\pi/2} (a+b) \cdot 1 \cdot dx = (a+b) \frac{\pi}{2} \quad \therefore I = \frac{\pi(a+b)}{4}$$

Ans: (a)

90. The value of $\int_0^2 \frac{dx}{\sqrt{4+4x-x^2}}$ is

Options:

(a) π

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) $\frac{\pi}{3}$

$$\text{Sol: } \int_0^2 \frac{dx}{\sqrt{-(x^2 - 4x + 4 - 8)}} = \int_0^2 \frac{dx}{\sqrt{8 - (x-2)^2}}$$

$$= \sin^{-1}(0) - \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = 0 + \frac{\pi}{4} = \frac{\pi}{4} \quad \text{Ans: (b)}$$

91. The system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ has no solution, if

Options:

- (a) $\lambda = 3, \mu = 10$
- (b) $\lambda \neq 3, \mu = 10$
- (c) $\lambda \neq 3, \mu \neq 10$
- (d) $\lambda = 3, \mu \neq 10$

Sol: Given, $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$

For no solution,

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 1(2\lambda - 6) - 1(\lambda - 3) + 1(2 - 2) = 0$$

$$\Rightarrow \lambda - 3 = 0 \Rightarrow \lambda = 3$$

$$\text{And } \Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ 10 & 2 & 3 \\ \mu & 2 & 3 \end{vmatrix} \neq 0$$

$$\Rightarrow 6(6 - 6) - 1(30 - 3\mu) + 1(20 - 2\mu) \neq 0$$

$$\Rightarrow \mu - 10 \neq 0 \text{ or } \mu \neq 10$$

Ans: (d)

92. If A is a square matrix of order 3 and $|A| = 3$, then $|\text{adj } A|$ is

Options:

- (a) 3
- (b) 9
- (c) (1, 3)
- (d) 0

Sol: If A is square matrix, of order n , and $|A| \neq 0$,

$$\text{Then } |\text{adj } A| = |A|^{n-1} = 3^{3-1} = 3^2 = 9$$

Ans: (b)

93. The value of $\tan^{-1} \left(\frac{\cos 90^\circ - \sin 90^\circ}{\cos 90^\circ + \sin 90^\circ} \right)$ is

Options:

- (a) $\frac{\pi}{3}$
- (b) $\frac{\pi}{4}$
- (c) $\frac{-\pi}{4}$
- (d) $\frac{\pi}{2}$

$$\text{Sol: Given expression} = \tan^{-1} \left(\frac{0 - 1}{0 + 1} \right) = \tan^{-1}(-1) = -\frac{\pi}{4}.$$

Ans: (c)

94. The value of $\sin \left[2 \cos^{-1} \left(\frac{3}{5} \right) \right]$ is

Options:

(a) $\frac{24}{25}$

(b) $\frac{2\sqrt{6}}{5}$

(c) $\frac{2\sqrt{5}}{6}$

(d) $\frac{16}{25}$

Sol: $\sin \left[2 \cos^{-1} \left[\frac{3}{5} \right] \right] = \sin \left[\cos^{-1} \left(2 \cdot \left(\frac{3}{5} \right)^2 - 1 \right) \right]$

$$= \sin \left[\cos^{-1} \left(\frac{18}{25} - 1 \right) \right]$$

$$= \sin \left[\cos^{-1} \left(-\frac{7}{25} \right) \right]$$

$$= \sin \left[\sin^{-1} \frac{24}{25} \right] = \frac{24}{25}.$$

Ans: (a)

95. If $\vec{p} = (2, 1, 3)$, $\vec{q} = (-2, 3, 1)$ and $\vec{r} = (3, -2, 4)$ and $\hat{j}$ is the unit vector in the direction of y -axis, then

$$(2\vec{p} + 3\vec{q} - 4\vec{r}) \cdot \hat{j} \text{ is}$$

Options:

(a) 18

(b) 19

(c) 20

(d) 21

Sol: $(2\vec{p} + 3\vec{q} - 4\vec{r}) \cdot \hat{j}$. As $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = 0$, consider only

$\hat{j}$ components

$$\therefore (2\hat{j} + 9\hat{j} + 8\hat{j}) \cdot \hat{j} = 19$$

Ans: (b)

96. The area of the parallelogram whose adjacent sides are $\hat{i} + \hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$ is

Options:

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) 3

(d) 4

Sol: Area of the parallelogram whose adjacent sides are $\vec{a}$ and $\vec{b}$ is $|\vec{a} \times \vec{b}|$

$$|\vec{a} \times \vec{b}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = \hat{i}(0-1) - \hat{j}(1-2) + \hat{k}(1-0)$$

$$= \hat{i} + \hat{j} + \hat{k} \therefore |\vec{a} \times \vec{b}| = \sqrt{1+1+1} = \sqrt{3}.$$

Ans: (b)

97. When direction cosines of two lines satisfy $l + m + n = 0$ and $2lm + 2ln = mn$, then (l, m, n) is equal to

Options:

(a) $(1, 2, 1)$

(b) $(2, -1, 1)$

(c) $\left(\frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$

(d) $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Sol: $\left(\frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ Satisfy both the conditions.

Ans: (c)

98. For $k = 1, 2, 3$ the box B_k contains k red balls and $(k+1)$ white balls. Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$

and $P(B_3) = \frac{1}{6}$. A box is selected at random and a ball is drawn from it. If a red balls is drawn, then the

probability that it has come from box B_2 , is

Options:

(a) $\frac{35}{78}$

(b) $\frac{14}{39}$

(c) $\frac{10}{13}$

(d) $\frac{12}{13}$

Sol: For $k = 1, 2, 3$ the box B_k contains k red balls and $(k+1)$ white balls.

Let $P(B_1) = \frac{1}{2}$, $P(B_2) = \frac{1}{3}$ and $P(B_3) = \frac{1}{6}$.

A box is selected at random and a ball is drawn from it.

$$P\left(\frac{B_2}{R}\right) = \frac{P(B_2)P\left(\frac{R}{B_2}\right)}{P(B_1)P\left(\frac{R}{B_1}\right) + P(B_2)P\left(\frac{R}{B_2}\right) + P(B_3)P\left(\frac{R}{B_3}\right)}$$

$$= \frac{\frac{1}{3} \times \frac{2}{5}}{\frac{1}{2} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{6} \times \frac{3}{7}} = \frac{\frac{2}{15}}{\frac{1}{6} + \frac{2}{15} + \frac{1}{14}} = \frac{14}{39}$$

Ans: (b)

99. Events A, B, C are mutually exclusive events such that $P(A) = \frac{3x+1}{3}$, $P(B) = \frac{1-x}{4}$ and $P(C) = \frac{1-2x}{2}$.

The set of possible values of x are in the interval

Options:

- (a) $\left[\frac{1}{3}, \frac{1}{2}\right]$
 (b) $\left[\frac{1}{3}, \frac{2}{3}\right]$
 (c) $\left[\frac{1}{3}, \frac{13}{3}\right]$
 (d) $[0, 1]$

Sol: Since, $0 \leq P(A) \leq 1, 0 \leq P(B) \leq 1, 0 \leq P(C) \leq 1$

and $0 \leq P(A) + P(B) + P(C) \leq 1$

$$\therefore 0 \leq \frac{3x+1}{3} \leq 1$$

$$\Rightarrow -\frac{1}{3} \leq x \leq \frac{2}{3} \quad \dots (i)$$

$$0 \leq \frac{1-x}{4} \leq 1 \Rightarrow -3 \leq x \leq 1 \quad \dots (ii)$$

$$0 \leq \frac{1-2x}{2} \leq 1 \Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2} \quad \dots (iii)$$

$$\text{and } 0 \leq \frac{3x+1}{3} + \frac{1-x}{4} + \frac{1-2x}{2} \leq 1 \Rightarrow \frac{1}{3} \leq x \leq \frac{13}{3} \quad \dots (iv)$$

From Eqs. (i), (ii), (iii) and (iv), we get $\frac{1}{3} \leq x \leq \frac{1}{2}$

Ans: (a)

100. A box contains 12 bulbs of which 3 are defective. If three bulbs are chosen at random without replacement, find the probability of at least one bulb is good.

Options:

- (a) $\frac{1}{120}$

(b) $\frac{21}{40}$

(c) $\frac{219}{220}$

(d) $\frac{1}{220}$

Sol: $P(\bar{B}) = 1 - P(B) = 1 - \frac{{}^3C_3}{{}^{12}C_3} = 1 - \frac{3 \times 2 \times 1}{12 \times 11 \times 10} = 1 - \frac{1}{220} = \frac{219}{220}$.

Ans: (c)

101. The number of terms in the expansion of $(a+b+c)^{10}$ is

Options:

(a) 11

(b) 21

(c) 55

(d) 66

Sol: The number of terms in the expansion of

$$(a+b+c)^{10} = {}^{12}C_2 = \frac{11 \cdot 12}{2} = 66$$

Ans: (d)

102. If x and y are acute angles, such that $\cos x + \cos y = \frac{3}{2}$ and $\sin x + \sin y = \frac{3}{4}$, then $\sin(x+y)$ equals

Options:

(a) $\frac{2}{5}$

(b) $\frac{3}{4}$

(c) $\frac{3}{5}$

(d) $\frac{4}{5}$

Sol: Given, $\cos x + \cos y = \frac{3}{2}$

$$\Rightarrow 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = \frac{3}{2} \quad \dots (i)$$

And $\sin x + \sin y = \frac{3}{4}$

$$\Rightarrow 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) = \frac{3}{4} \quad \dots (ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)}{2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right)} = \frac{\frac{3}{4}}{\frac{3}{2}} \Rightarrow \tan\left(\frac{x+y}{2}\right) = \frac{1}{2}$$

$$\therefore \sin(x+y) = \frac{2 \tan\left(\frac{x+y}{2}\right)}{1 + \tan^2\left(\frac{x+y}{2}\right)} = \frac{2 \times \frac{1}{2}}{1 + \left(\frac{1}{2}\right)^2} = \frac{4}{4+1} = \frac{4}{5}$$

Ans: (d)

103. $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 180^\circ$ is equal to

Options:

- (a) 1
- (b) 0
- (c) 2
- (d) -1

$$\begin{aligned} \text{Sol: } & \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 89^\circ + \cos 90^\circ + \cos 91^\circ + \cos 92^\circ + \cos 93^\circ + \dots + \cos 179^\circ + \cos 180^\circ \\ &= \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 89^\circ + 0 + \cos(180^\circ - 89^\circ) + \cos(180^\circ - 88^\circ) + \dots + \cos(180^\circ - 1^\circ) - 1 \\ &= \cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 89^\circ - \cos 89^\circ - \cos 88^\circ - \dots - \cos 1^\circ - 1 = -1 \end{aligned}$$

Ans: (d)

104. The value of k such that the lines $2x - 3y + k = 0$, $3x - 4y - 13 = 0$ and $8x - 11y - 33 = 0$ are concurrent, is

Options:

- (a) 20
- (b) -7
- (c) 7
- (d) -20

Sol: Since, given lines are concurrent

$$\therefore \begin{vmatrix} 2 & -3 & k \\ 3 & -4 & -13 \\ 8 & -11 & -33 \end{vmatrix} = 0$$

$$\Rightarrow 2(132 - 143) + 3(-99 + 104) + k(-33 + 32) = 0$$

$$\Rightarrow -22 + 15 - k = 0 \Rightarrow k = -7$$

Ans: (b)

105. A straight line passes through the points $(5, 0)$ and $(0, 3)$. The length of perpendicular from the point $(4, 4)$ on the line is

Options:

- (a) $\frac{15}{\sqrt{34}}$
- (b) $\frac{\sqrt{17}}{2}$

(c) $\frac{17}{2}$

(d) $\sqrt{\frac{17}{2}}$

Sol: The equation of line passing through points (5, 0) and (0, 3) is

$$(y-0) = \frac{(3-0)}{(0-5)}(x-5) \Rightarrow y = \frac{3}{-5}(x-5)$$

$$\Rightarrow -5y = 3x - 15 \Rightarrow 3x + 5y - 15 = 0$$

So, the length of perpendicular from the point (4, 4) to the above line,

$$d = \frac{(3 \times 4 + 5 \times 4 - 15)}{\sqrt{3^2 + 5^2}} = \frac{12 + 20 - 15}{\sqrt{9 + 25}}$$

$$= \frac{17}{\sqrt{34}} = \sqrt{\frac{17}{2}}$$

Ans: (d)

106. If one end of a diameter of the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ is (3, 4), then find the coordinates of the other end of the diameter

Options:

(a) (2, 1)

(b) (1, 2)

(c) (1, 1)

(d) None of these

Sol: The centre of the given circle $x^2 + y^2 - 4x - 6y + 11 = 0$ is (2, 3)

Let the other end of diameter be (x_1, y_1)

Since, centre = $\left(\frac{x_1 + 3}{2}, \frac{y_1 + 4}{2}\right)$

$$\therefore (2, 3) = \left(\frac{x_1 + 3}{2}, \frac{y_1 + 4}{2}\right) \Rightarrow 2 = \frac{x_1 + 3}{2}, 3 = \frac{y_1 + 4}{2}$$

$$\therefore x_1 = 1, y_1 = 2$$

Hence, coordinates of other end of the diameter is (1, 2)

Ans: (b)

107. The equation of a directrix of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ is

Options:

(a) $3y = \pm 5$

(b) $y = \pm 5$

(c) $3y = \pm 25$

(d) $y = \pm 3$

Sol: Here, $a = 4$, $b = 5$ and $e = \sqrt{1 - \frac{16}{25}} = \frac{3}{5}$

$\therefore$ Equation of directrix is $y = \pm \frac{b}{e} \Rightarrow y = \pm \left(\frac{5}{3/5} \right)$

$\Rightarrow 3y = \pm 25$

Ans: (c)

108. The eccentricity of the hyperbola with latus rectum 12 and semi-conjugate axis $2\sqrt{3}$ is

Options:

(a) 3

(b) $\sqrt{\frac{3}{2}}$

(c) $2\sqrt{3}$

(d) 2

Sol: Since, $\frac{2b^2}{a} = 12$ and $b = 2\sqrt{3}$

$\therefore a = \frac{b^2}{6} = 2$

$\therefore e = \sqrt{\frac{a^2 + b^2}{a^2}} = \sqrt{\frac{4 + 12}{4}} = 2$

Ans: (d)

109. The sum of the reciprocals of focal distances of a focal chord PQ of $y^2 = 4ax$ is

Options:

(a) $\frac{1}{a}$

(b) a

(c) $2a$

(d) $\frac{1}{2a}$

Sol: Required sum $= \frac{1}{2a} + \frac{1}{2a} = \frac{2}{2a} = \frac{1}{a}$

Ans: (a)

110. A spherical balloon is being inflated at the rate of 35 cc/min. The rate of increase of the surface area of the balloon when its diameter is 14 cm is

Options:

(a) 7 sq.cm/min

(b) 10 sq.cm/min

(c) 17.5 cm/min

(d) 28 sq.cm/min

Sol: By data $\frac{dV}{dt} = 35 \text{ cc / min}$

$$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$\Rightarrow \frac{35}{4\pi r^2} = \frac{dr}{dt}$$

$$\Rightarrow \left(\frac{dr}{dt}\right)_{r=7} = \frac{5}{28\pi} \text{ cm / min}$$

$$S = 4\pi r^2 \Rightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$\Rightarrow \left(\frac{dS}{dt}\right)_{r=7} = 8 \cdot \pi \cdot 7 \cdot \frac{5}{28\pi}$$

$$= 10 \text{ sq.cm / min}$$

Ans: (b)

111. If $a < 0$, the function $f(x) = e^{ax} + e^{-ax}$ is decreasing for all values of x , where

Options:

(a) $x > 0$

(b) $x < 0$

(c) $x > 1$

(d) $x < 1$

$$\text{Sol: } f(x) = e^{ax} + e^{-ax} \Rightarrow f'(x) = ae^{ax} - ae^{-ax}$$

$$\text{Now, } f'(x) < 0 \Rightarrow a(e^{ax} - e^{-ax}) < 0$$

$$\Rightarrow e^{ax} - e^{-ax} > 0 \quad (\because a < 0)$$

$$\Rightarrow e^{2ax} - 1 > 0$$

$$\Rightarrow e^{2ax} > 1$$

$$\Rightarrow e^{2ax} > e^0$$

$$\Rightarrow ax > 0$$

$$\Rightarrow x < 0 \quad (\because a < 0)$$

Ans: (b)

112. On the interval $[0, 1]$ the function $x^{25}(1-x)^{75}$ takes its maximum value at the point

Options:

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

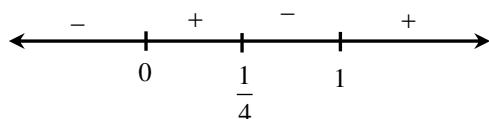
(d) $\frac{1}{3}$

Sol: We have, $f(x) = x^{25}(1-x)^{75}$

$$\Rightarrow f'(x) = 25x^{24}(1-x)^{75} - 75x^{25}(1-x)^{74}$$

$$\Rightarrow f'(x) = 25x^{24}(1-x)^{74}(1-4x)$$

The sign of $f'(x)$, is shown below:



Clearly $f'(x)$ changes from +ve to -ve at $x = \frac{1}{4}$

$\therefore f(x)$ attains maximum at $x = \frac{1}{4}$

Ans: (b)

113. The sum of an infinite geometric series is 2 and the sum of the geometric series made from the cubes of this infinite series is 24. Then the series is

Options:

(a) $3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \dots$

(b) $3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \dots$

(c) $3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \dots$

(d) none of these

Sol: Let first term = a , common ratio = r ,

Where $-1 < r < 1$

Then, $\frac{a}{1-r} = 2$ and $\frac{a^3}{1-r^3} = 24$

$$\therefore \frac{1-r^3}{(1-r)^3} = \frac{1}{3}$$

i.e. $1 - 2r + r^2 = 3(1 + r + r^2)$

$$2r^2 + 5r + 2 = 0$$

$$\therefore r = -2 \text{ or } -\frac{1}{2}$$

As $-1 < r < 1$ $\therefore$ We have $r = -\frac{1}{2}$

Putting this value of r , we get $a = 3$,

∴ The series is $3 - \frac{3}{2} + \frac{3}{4} - \frac{3}{8} + \dots$

Hence (c) is the correct answer

Ans: (c)

114. The minimum value of $4^x + 4^{1-x}$, $x \in R$, is

Options:

- (a) 2
- (b) 4
- (c) 1
- (d) none of these

Sol: A.M $\geq$ G.M for positive numbers

$$\text{So, } \frac{4^x + \frac{4}{4^x}}{2} \geq \sqrt{4^x \cdot \frac{4}{4^x}} = 2 \Rightarrow 4^x + 4^{1-x} \geq 4$$

Hence (b) is the correct answer

Ans: (b)

115. ${}^{15}C_8 + {}^{15}C_9 - {}^{15}C_6 - {}^{15}C_7$ is equal to

Options:

- (a) 8
- (b) 0
- (c) 6
- (d) none of these

Sol: We have,

$$\begin{aligned} {}^{15}C_8 + {}^{15}C_9 - {}^{15}C_6 - {}^{15}C_7 &= ({}^{15}C_8 + {}^{15}C_9) - ({}^{15}C_6 + {}^{15}C_7) \\ &= {}^{16}C_9 - {}^{16}C_7 \quad \left(\because {}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1} \right) \\ &= {}^{16}C_9 - {}^{16}C_9 \quad \left(\because {}^nC_r = {}^nC_{n-r} \right) \\ &= 0 \end{aligned}$$

Hence (b) is correct answer

Ans: (b)

116. The smallest positive integral value of n for which $\left(\frac{1-i}{1+i}\right)^n$ is purely imaginary with positive imaginary part, is

Options:

- (a) 1
- (b) 3
- (c) 5
- (d) none of these

Sol: $\frac{1-i}{1+i} = \frac{(1-i)^2}{1-i^2} = \frac{-2i}{2} = -i$

$\therefore \left(\frac{1-i}{1+i}\right)^n = (-i)^n = \text{imaginary} \Rightarrow n = 1, 3, 5, \dots$

Which is +ve imaginary for $n = 3, 7, \dots$

$\therefore n = 3$ is smallest positive integer

Hence (b) is correct answer

Ans: (b)

117. The conjugate complex number of $\frac{2-i}{(1-2i)^2}$ is

Options:

(a) $\frac{2}{25} + \frac{11}{25}i$

(b) $\frac{2}{25} - \frac{11}{25}i$

(c) $-\frac{2}{25} + \frac{11}{25}i$

(d) $-\frac{2}{25} - \frac{11}{25}i$

Sol: Let $z = \frac{2-i}{(1-2i)^2}$

$\therefore z = \frac{2-i}{-3-4i} = \frac{-2+i}{3+4i} \times \frac{3-4i}{3-4i}$

$= -\frac{2}{25} + \frac{11}{25}i$

$\therefore \bar{z} = -\frac{2}{25} - \frac{11}{25}i$

Ans: (d)

118. The multiplicative inverse of $(6+5i)^2$ is

Options:

(a) $\frac{11}{61} - \frac{60}{61}i$

(b) $\frac{11}{61} + \frac{60}{61}i$

(c) $\frac{9}{61} - \frac{60}{61}i$

(d) none of these

Sol: Let $z = (6+5i)^2 = 36 + 2(6)(5i) + 25i^2$

$= 36 + 60i - 25 = 11 + 60i$

Then $\bar{z} = 11 - 60i$

And $|z| = \sqrt{(11)^2 + (60)^2} = \sqrt{121 + 3600} = \sqrt{3721} = 61$

Multiplicative inverse of

$$z = \frac{\bar{z}}{|z|^2} = \frac{11 - 60i}{(61)^2} = \frac{11}{(61)^2} - \frac{60}{(61)^2}i$$

Ans: (d)

119. The standard deviation for the set of numbers 1, 4, 5, 7, 8 is 2.45 nearly. If 10 is added to each number then, new standard deviation is

Options:

- (a) 24.45
- (b) 12.45
- (c) 2.45
- (d) 0.245

Sol: The standard deviation remains same = 2.45

Ans: (c)

120. The mean of 18 observations is 7 and if each observation 5 is added, then the new mean will be

Options:

- (a) 2
- (b) 12
- (c) 7
- (d) none of these

Sol: New mean = $7 + 5 = 12$

Ans: (b)

Physics

Multiple Choice Questions with one correct answer. A correct answer carries 1 mark. No negative mark. **60 x 1 = 60**

121. Out of the following pairs which one does not have identical dimensions?

- (a) Moment of inertia and moment of a force
- (b) Work and torque
- (c) Angular momentum and Planck's constant
- (d) Impulse and momentum

Sol: Moment of inertia $(I) = mr^2$

$$\therefore [I] = [ML]^2$$

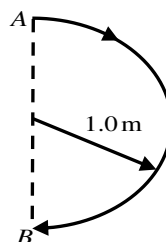
Moment of force $(\vec{\tau}) = \vec{r} \times \vec{F}$

$$\therefore [\tau] = [r][F] = [L][MLT^{-2}]$$

or $[\tau] = [ML^2T^{-2}]$

Moment of inertia and moment of a force do not have identical dimensions

Ans: (a)



122. In 1.0 s, a particle goes from point A to point B, moving in a semicircle of radius 1.0 m. The magnitude of the average velocity of the particle is

- (a) 3.14 m s^{-1}
- (b) 2.0 m s^{-1}
- (c) 1.0 m s^{-1}
- (d) zero

Sol: Average velocity = $\frac{\text{Displacement}}{\text{time}}$

$$= \frac{AB}{t} = \frac{2.0 \text{ m}}{1.0 \text{ s}} = 2.0 \text{ m s}^{-1}$$

Ans: (b)

123. If wavelength of maximum intensity of radiation emitted by sun and moon are $0.5 \times 10^{-6} \text{ m}$ and 10^{-4} m respectively, the ratio of the temperature is

- (a) 1 : 100
- (b) 1 : 200
- (c) 200 : 1
- (d) 400 : 1

Sol:

According to Wein's displacement law,

$$\lambda_m T = \text{constant}$$

$$\therefore \frac{T_1}{T_2} = \frac{\lambda_{m2}}{\lambda_{m1}} = \frac{10^{-4}}{0.5 \times 10^{-6}} = 200$$

Ans: (c)

124. A body is allowed to fall from a height of 100 m. If the time taken for the first 50 m is t_1 , and for the remaining 50 m is t_2 , then

- (a) $t_1 = t_2$
- (b) $t_1 > t_2$
- (c) $t_1 < t_2$
- (d) depends upon the mass

Sol: $s = \frac{1}{2}gt_1^2$

or $t_1^2 = \frac{50 \times 2}{g} = \frac{100}{g}$ or $t_1 = \frac{10}{\sqrt{g}}$

and $100 = \frac{1}{2}gt^2$ or $t = \frac{10\sqrt{2}}{\sqrt{g}}$

$\therefore t_2 = t - t_1 = \frac{10}{\sqrt{g}}(\sqrt{2} - 1) = 0.4t_1$ i.e., $t_1 > t_2$

Ans: (b)

125. A wall is made of equally thick layers A and B of different materials. Thermal conductivity of A is twice that of B . In the steady state, the temperature difference across the wall is 36°C . The temperature difference across the layer A is

(a) 12°C

(b) 18°C

(c) 6°C

(d) 24°C

Sol: Here, $K_A = 2K_B$, $\theta_A - \theta_B = 36^\circ\text{C}$

Let θ is the temperature of the junction.

As $\left(\frac{\Delta\theta}{\Delta t}\right)_A = \left(\frac{\Delta\theta}{\Delta t}\right)_B$

$\therefore \frac{K_A A (\theta_A - \theta)}{x} = \frac{K_B A (\theta - \theta_B)}{x}$

$2K_B (\theta_A - \theta) = K_B (\theta - \theta_B)$

$2(\theta_A - \theta) = \theta - \theta_B$

Add $(\theta_A - \theta)$ on both sides, we get

$3(\theta_A - \theta) = \theta_A - \theta + \theta - \theta_B$

$3(\theta_A - \theta) = \theta_A - \theta_B$

$\theta_A - \theta = \frac{\theta_A - \theta_B}{3} = \frac{36}{3} = 12^\circ\text{C}$

$\therefore$ Temperature difference across the layer A

$= \theta_A - \theta = 12^\circ\text{C}$

Ans: (a)

126. Two bodies are thrown up at angles of 45° and 60° respectively, with the horizontal. If both bodies attain same vertical height, then the ratio of velocities with which these are thrown is

(a) $\sqrt{\frac{2}{3}}$

(b) $\frac{2}{\sqrt{3}}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\frac{\sqrt{3}}{2}$

Sol: Vertical height $= \frac{u^2 \sin^2 \theta}{2g}$

where u is initial velocity of the body

$\therefore \frac{u_1^2 \sin^2 45^\circ}{2g} = \frac{u_2^2 \sin^2 60^\circ}{2g}$

$$\therefore \frac{u_1^2}{u_2^2} = \frac{\sin^2 60^\circ}{\sin^2 45^\circ} = \frac{3/4}{1/2} = \frac{3}{2}$$

$$\therefore \frac{u_1}{u_2} = \sqrt{\frac{3}{2}}$$

Ans: (c)

127. 400 cc volume of a gas having $\gamma = 5/2$ is suddenly compressed to 100 cc. If the initial pressure is P , then the final pressure will be

- (a) $P/32$ (b) $8P$ (c) $32P$ (d) $16P$

Sol: Here $V_1 = 400\text{cc}$, $V_2 = 100\text{cc}$; $\gamma = \frac{5}{2}$

$$P_1 = P, P_2 = ?$$

Sudden compression is an adiabatic process.

For an adiabatic process, $PV^\gamma = \text{constant}$

$$\text{As } P_2 V_2^\gamma = P_1 V_1^\gamma$$

$$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = P \left(\frac{400}{100} \right)^{5/2} = P \times 4 \times 4\sqrt{4} = 32P$$

Ans: (c)

128. A spring of force constant 800 N m^{-1} has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is

- (a) 16 J (b) 8 J (c) 32 J (d) 24 J

$$\text{Sol: } W = \frac{1}{2} k (x_2^2 - x_1^2)$$

$$W = \frac{1}{2} \times 800 [(15 \times 10^{-2})^2 - (5 \times 10^{-2})^2]$$

$$W = 400 \times 200 \times 10^{-4} = 8 \text{ J}$$

Ans: (b)

129. If the air column in a pipe which is closed at one end, is in resonance with a vibrating tuning fork at a frequency 260 Hz, then the length of the air column is

- (a) 35.7 cm (b) 31.7 cm (c) 12.5 cm (d) 62.5 cm

Sol: Frequency of the tuning fork, $\nu = 260 \text{ Hz}$.

The fundamental frequency of a closed organ pipe,

$$\nu = \frac{v}{4l}$$

where v is the velocity of sound in air

$$\text{Therefore length of air column } (l) = \frac{v}{4(\nu)} = \frac{330}{4 \times 260}$$

$$= 0.317 \text{ m} = 31.7 \text{ cm}$$

Ans: (b)

130. Which of the following is an expression for power?

- (a) $P = \tau\omega$ (b) $P = I\alpha$ (c) $P = I\omega$ (d) $P = \tau\alpha$

Sol: $P = \tau\omega$. ($P = FV$ in translational motion)

Since $\tau = I\alpha$

$$\therefore P = I\omega\alpha$$

Ans: (a)

131. A body constrained to move in y - direction is subjected to a force given by $\vec{F} = (-2\hat{i} + 15\hat{j} + 6\hat{k})\text{N}$.

The work done by this force in moving the body a distance of 10 m along the y -axis is

- (a) 20 J (b) 150 J (c) 160 J (d) 190 J

Sol: Given : $\vec{F} = -2\hat{i} + 15\hat{j} + 6\hat{k} \text{ N}$

$$\vec{S} = 0\hat{i} + 10\hat{j} + 0\hat{k} \text{ m}$$

$\therefore$ work done

$$\vec{F} \cdot \vec{S} = (-2\hat{i} + 15\hat{j} + 6\hat{k}) \cdot (0\hat{i} + 10\hat{j} + 0\hat{k}) = 150 \text{ J}$$

Ans: (a)

132. A particle performing uniform circular motion has angular momentum L . If its angular frequency is doubled and its kinetic energy halved, then the new angular momentum is

- (a) $\frac{L}{4}$ (b) $2L$ (c) $4L$ (d) $\frac{L}{2}$

Sol: Angular momentum ... (i)

$$L = I\omega$$

Kinetic energy,

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}L\omega \quad [\text{From (i)}]$$

$$\therefore L = \frac{2K}{\omega}$$

$$\text{Now, } L' = \frac{2\left(\frac{K}{2}\right)}{2\omega} \Rightarrow L' = \frac{L}{4}$$

Ans: (c)

133. If the metal bob of a simple pendulum is replaced by a wooden bob, then its time period will

- (a) increase (b) decrease
(c) remain the same (d) first increases then decreases

Sol: Time period of a simple pendulum is

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Time period of a simple pendulum is independent on the material of the bob, Hence when a metal bob is replaced by a wooden bob its time period remains the same.

Ans: (c)

134. Two satellites of masses M and $9M$ are orbiting a planet in a circular orbit of radius r . Their frequency of revolution will be in the ratio of

- (a) 1 : 9 (b) 1 : 3 (c) 1 : 1 (d) 3 : 1

Sol: Time period or frequency of satellite is independent of mass: $T^2 \propto r^3$

$$\Rightarrow \frac{f_1}{f_2} = 1$$

Ans: (c)

135. Two tuning forks A and B vibrating simultaneously produce 5 beats. Frequency of B is 512 Hz. It is seen that if one arm of A is filed, then the number of beats increases. Frequency of A will be

- (a) 502 Hz (b) 507 Hz (c) 517 Hz (d) 522 Hz

Sol: There are five beats between A and B , therefore, the possible frequencies of A are $512 \pm 5 = (517 \text{ or } 507) \text{ Hz}$.

When one prong of A is filed its frequency becomes greater than the original frequency. If we assume that the original frequency of A is 517 Hz. The beats between A and B will be more than 5. But it is given that the beats are increasing so it is only possible if frequency of A is 517 Hz.

Ans: (c)

136. Two point charges exert on each other a force F when they are placed r distance apart in air. When they are placed R distance apart in medium of dielectric constant K , they exert the same force. The distance R equals

- (a) $\frac{r}{\sqrt{K}}$ (b) $\frac{r}{K}$ (c) rK (d) $r\sqrt{K}$

$$\text{Sol: } F_1 = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} \text{ or } F_2 = \frac{1}{4\pi\epsilon_0 K} \frac{q^2}{R^2}$$

$$\text{As } F_1 = F_2$$

$$\text{hence } \frac{q^2}{r^2} = \frac{1}{K} \frac{q^2}{R^2} \quad \therefore \quad R = r / \sqrt{K}$$

Ans: (a)

137. If potential difference of a capacitor $6\mu\text{F}$ is changed from 10 V to 20 V, then increase in energy is

- (a) $2 \times 10^{-4} \text{ J}$ (b) $4 \times 10^{-4} \text{ J}$ (c) $3 \times 10^{-4} \text{ J}$ (d) $9 \times 10^{-4} \text{ J}$

Sol: Increase in energy

$$= U_2 - U_1 = \frac{1}{2} C (V_2^2 - V_1^2)$$

$$= \frac{1}{2} \times 6 \times 10^{-6} (20^2 - 10^2) = 9 \times 10^{-4} \text{ J}$$

Ans: (d)

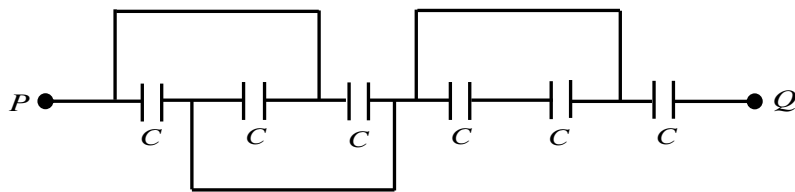
138. The electric field E , current density J and conductivity σ of a conductor are related as

- (a) $\sigma = E / J$ (b) $\sigma = J / E$ (c) $\sigma = J^2 E$ (d) $\sigma = \frac{1}{JE}$

Sol: $J = \sigma E$ or $\sigma = J / E$

Ans: (b)

139. For circuit the equivalent capacitance between points P and Q is



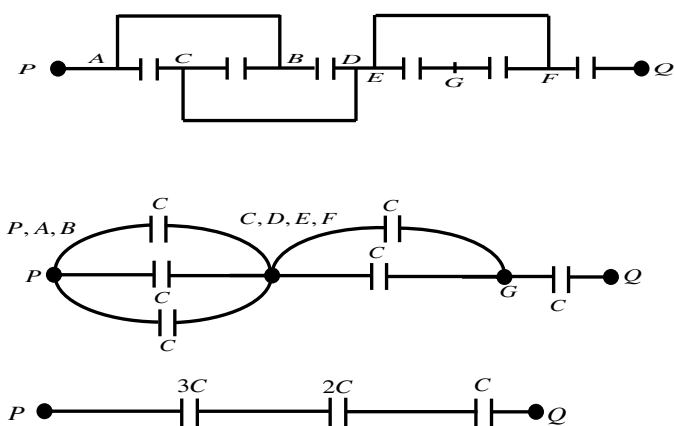
(a) $6C$

(b) $4C$

(c) $\frac{3}{2}C$

(d) $\frac{6}{11}C$

Sol:



$$\therefore \frac{1}{C_{eq}} = \frac{1}{3C} + \frac{1}{2C} + \frac{1}{C} = \frac{2+3+6}{6C} = \frac{11}{6C}$$

$$\therefore C_{eq} = \frac{6C}{11}$$

Ans: (d)

140. A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10 V. The potential at the centre of the sphere is

(a) 0 V

(b) 10 V

(c) same as at a point 5 cm away from the surface

(d) same as at a point 20 cm away from the surface

Sol: Potential at the centre = potential on the surface = 10 V

Ans: (b)

141. A negatively charged oil drop is prevented from falling under gravity by applying a vertical electric field

100 V m^{-1} . If the mass of the drop is $1.6 \times 10^{-3} \text{ g}$, the number of electron carried by the drop is

($g = 10 \text{ m s}^{-2}$)

(a) 10^{18}

(b) 10^{15}

(c) 10^{12}

(d) 10^9

Sol: For the drop to be stationary

Force on the drop due to electric field = Weight of the drop $qE = mg$

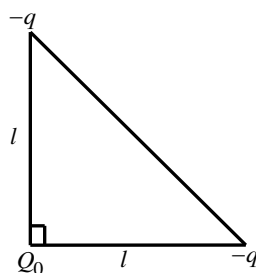
$$q = \frac{1.6 \times 10^{-6} \times 10}{100} = 1.6 \times 10^{-7} C$$

Number of electrons carried by the drop is

$$= \frac{q}{e} = \frac{1.6 \times 10^{-7} C}{1.6 \times 10^{-19} C} = 10^{12}$$

Ans: (c)

142. Three charges $Q_0, -q$ and $-q$ are placed at the vertices of an isosceles right angled triangle as in the figure. The net electrostatic potential energy is zero if Q_0 is equal to



(a) $\frac{q}{4}$

(b) $\frac{2q}{\sqrt{32}}$

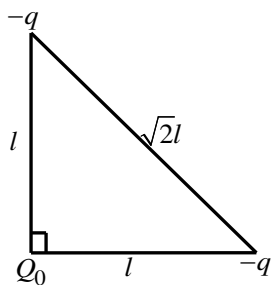
(c) $\sqrt{2}q$

(d) $+q$

Sol: The net potential energy = 0 (given)

$$U_{\text{Total}} = \frac{1}{4\pi\epsilon_0} \frac{q^2}{\sqrt{2}l} - \left(\frac{1}{4\pi\epsilon_0} \right) \frac{2qQ_0}{l} = 0$$

$$\therefore Q_0 = \frac{1}{2\sqrt{2}} \Rightarrow Q_0 = \frac{2q}{4\sqrt{2}} = \frac{2q}{\sqrt{32}}$$



Ans: (b)

143. The potential gradient along the length of a uniform wire is 10 volt per metre. B and C are two points at 30 cm and 60 cm on a metre scale along the wire. The potential difference between B and C will be

(a) 3 volt

(b) 0.4 volt

(c) 7 volt

(d) 4 volt

Sol: Distance between two points

$$= 60 - 30 = 30 \text{ cm} = 0.3 \text{ m}$$

The potential gradient = 10 V/m

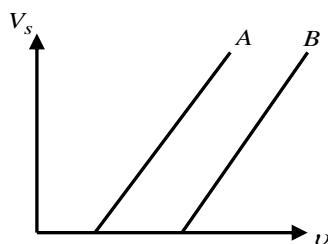
$\therefore$ Potential difference = potential gradient $\times$ length

$$= 10 \times 0.3 = 3V$$

Ans: (a)

144. The figure shows stopping potential V_s and frequency ν for two different metallic surface A and B .

The work function of A , as compared to that of B is

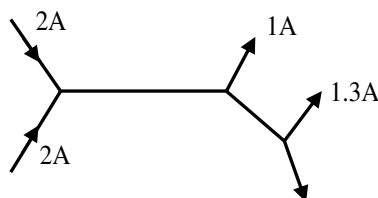


- (a) less (b) more (c) equal (d) nothing can be said

Sol: The value of threshold frequency ν_0 for A is less than that for B , hence $\phi_A < \phi_B$.

Ans: (a)

145. Figure shows currents in a part of an electric circuit, then current I is



- (a) 1.7 A (b) 3.7 A (c) 1.3 A (d) 1 A

Sol: Applying Kirchhoff's first law,

$$I = 2 + 2 - 1 - 1.3 = 1.7 A$$

Ans: (a)

146. The emf of a cell is ε and its internal resistance is r . Its terminals are connected to a resistance R . The potential difference between the terminals is 1.6 V for $R = 4\Omega$, and 1.8 V for $R = 9\Omega$. Then,

- (a) $\varepsilon = 1V, r = 1\Omega$ (b) $\varepsilon = 2V, r = 1\Omega$ (c) $\varepsilon = 2V, r = 2\Omega$ (d) $\varepsilon = 2.5V, r = 0.5\Omega$

Sol: Current in the circuit, $I = \frac{\varepsilon}{R + r}$

$$\text{p.d. across cell} = \text{p.d. across } R = IR = \frac{\varepsilon R}{R + r}$$

Set up two equations with the given data

$$1.6 = \frac{4\varepsilon}{4 + r} \quad \dots(i)$$

$$1.8 = \frac{9\varepsilon}{9 + r} \quad \dots(ii)$$

Solving equations (i) and (ii), we get

$$r = 1\Omega, \varepsilon = 2V$$

Ans: (b)

147. A wire of resistance R is stretched till its length is double of the original wire. Then the resistance of the stretched wire is

- (a) $2R$ (b) $4R$ (c) $8R$ (d) $16R$

Sol: Resistance R of a wire of length l and cross-section area A with specific resistance ρ for its materials is given by

$$R = \frac{\rho L}{A}$$

On stretching the volume remains constant

$$\therefore Al = A'l'$$

$$\text{or } A' = \frac{Al}{(2l)} = \frac{A}{2}$$

Let the resistance of the stretched wire be R' .

$$\therefore R' = \frac{\rho l'}{A'} = \frac{\rho(2l)}{(A/2)} = \frac{4\rho l}{A} = 4R$$

Ans: (b)

148. An electric heating element consumes 500 W when connected to a 100 V line. If the line voltage becomes 150 V, the power consumed will be

- (a) 500 W (b) 750 W (c) 1000 W (d) 1125 W

$$\text{Sol: Power, } P = \frac{V^2}{R}$$

As R is constant

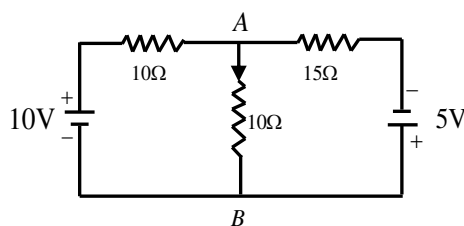
$$\therefore \frac{P_1}{P_2} = \left[\frac{V_1}{V_2} \right]^2 \text{ or } P_2 = P_1 \left(\frac{V_2}{V_1} \right)^2$$

Substituting the given values, we get

$$P_2 = 500 \times \left(\frac{150}{100} \right)^2 = 1125 \text{ W}$$

Ans: (d)

149. A circuit is as shown in the figure. Then, the current from A to B is



- (a) + 500 mA (b) + 250 mA (c) - 250 mA (d) - 500 mA

Sol: Applying Kirchhoff second law for a closed loop $CABDC$, we get

$$-10I_1 - 10(I_1 - I_2) + 10 = 0$$

$$-20I_1 + 10I_2 = -10$$

$$2I_1 - I_2 = 1 \quad \dots (i)$$

Again, applying Kirchhoff second law for a closed loop $ABFEA$, we get

$$-10(I_1 - I_2) - 5 + 15I_2 = 0$$

$$-10I_1 + 25I_2 = 5$$

$$2I_1 - 5I_2 = -1 \quad \dots(ii)$$

Solving (i) and (ii), we get

$$I_1 = \frac{3}{4}A, I_2 = \frac{1}{2}A$$

The current flows from A to B is

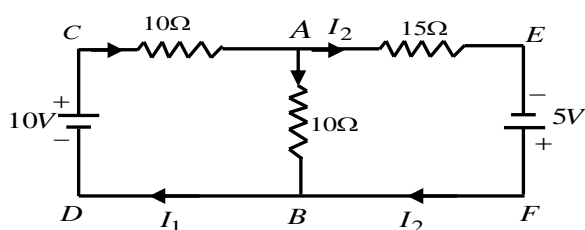
$$= I_1 - I_2 = \frac{3}{4}A - \frac{1}{2}A = \frac{1}{4}A = 0.25A = 250mA$$

Ans: (b)

150. The ratio of the de Broglie wavelength of proton and α particle which have been accelerated through same potential difference is

- (a) $2\sqrt{3}$ (b) $3\sqrt{2}$ (c) $2\sqrt{2}$ (d) $3\sqrt{3}$

Sol:



K.E. gained by a charge q after being accelerated through a potential difference V volt is given by

$$qV = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2qV}{m}} \text{ and } mv = \sqrt{2mqV}$$

$$\text{de Broglie wavelength, } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$$

$$\text{Now, } \lambda_p = \frac{h}{\sqrt{2m_p q_p V_p}}$$

For α particle

$$\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha q_\alpha V_\alpha}} \therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha V_\alpha}{m_p q_p V_p}}$$

Putting $V_\alpha = V_p$, we get

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{\frac{4 \times 2}{1 \times 1}} = \sqrt{8} = 2\sqrt{2}$$

Ans: (c)

151. The relative permeability of a medium is 0.075. what is its magnetic susceptibility?

- (a) 0.925 (b) - 0.925 (c) 1.075 (d) - 1.075

Sol: From $\mu_r = 1 + \chi_m$; $\chi_m = \mu_r - 1$

$$\chi_m = 0.075 - 1 = -0.925$$

Ans: (b)

152. If a galvanometer current is 10 mA, resistance of the galvanometer is 40Ω and shunt of 2Ω is connected to the galvanometer, the maximum current which can be measured by this ammeter is

- (a) 0.21 A (b) 2.1 A (c) 210 A (d) 21 A

$$\text{Sol: } I = \left(\frac{S+G}{S} \right) I_g = \left(\frac{2+40}{2} \right) \times 0.01 = 0.21 \text{ A}$$

Ans: (a)

153. A magnet of length 0.1 m and pole strength 10^{-4} Am is kept in a magnetic field of 30 Wb/m^2 at an angle of 30° . The couple acting on it is... $\times 10^{-4} \text{ Nm}$

- (a) 7.5 (b) 1.5 (c) 4.5 (d) 6.0

$$\text{Sol: Here, } 2l = 0.1 \text{ m, } m = 10^{-4} \text{ Am}$$

$$B = 30 \text{ Wb/m}^2, \theta = 30^\circ, \tau = ?$$

$$\tau = MB \sin \theta = m(2l)B \sin \theta$$

$$= 10^{-4} (0.1) \times 30 \times \sin 30^\circ = 1.5 \times 10^{-4} \text{ Nm.}$$

Ans: (b)

154. To produce a magnetic field of π tesla at the centre of circular loop of diameter 1 m, the current flowing through loop is

- (a) $5 \times 10^6 \text{ A}$ (b) 10^7 A (c) $2.5 \times 10^6 \text{ A}$ (d) $2 \times 10^6 \text{ A}$

$$\text{Sol: } B = \frac{\mu_0 2\pi I}{4\pi r}$$

$$\text{or } I = \frac{2Br}{\mu_0} = \frac{2 \times \pi \times 0.5}{4\pi \times 10^{-7}} = 2.5 \times 10^6 \text{ A}$$

Ans: (c)

155. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of magnetic moment is

- (a) $2 : \pi$ (b) $\pi : 2$ (c) $\pi : 4$ (d) $4 : \pi$

$$\text{Sol: Length of each side of square} = \frac{l}{4}$$

$$\therefore \text{Area of square, } A_1 = \left(\frac{l}{4} \right)^2 = \frac{l^2}{16}$$

$$\text{If } r \text{ is radius of the circle, then } l = 2\pi r, r = \frac{l}{2\pi}$$

$$\text{Area of circle, } A_2 = \pi r^2 = \pi \left(\frac{l}{2\pi} \right)^2 = \frac{l^2}{4\pi}$$

$$\frac{M_1}{M_2} = \frac{IA_1}{IA_2} = \frac{l^2/16}{l^2/4\pi} = \frac{4\pi}{16} = \frac{\pi}{4} = \pi : 4$$

Ans: (c)

156. Two long conductors, separated by a distance d carry current I_1 and I_2 in the same direction. They exert a force F on each other. Now the current in one of them is increased to two times and its direction is reversed. The distance is also increased to $3d$. The new value of the force between them is

- (a) $2F$ (b) $F/3$ (c) $2F/3$ (d) $F/3$

Sol: Initially, $F = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} l$

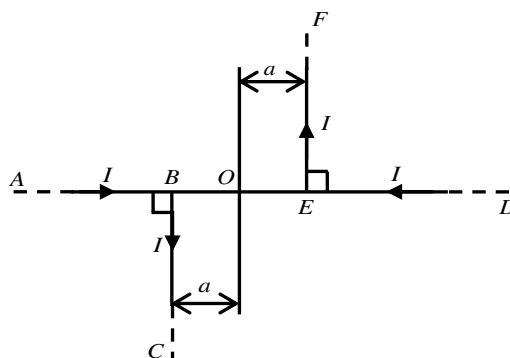
Finally, $F' = \frac{\mu_0}{2\pi} \frac{(2I_1)(I_2)}{3d} l$

$$\therefore \frac{F'}{F} = \frac{\mu_0}{2\pi} \frac{2I_1 I_2 l}{3d} \times \frac{2\pi d}{\mu_0 I_1 I_2 l} = \frac{2}{3}$$

$$\therefore F' = \frac{2}{3} F$$

Ans: (c)

157. Two long thin wires ABC and DEF are arranged as shown in figure. They carry currents I . The magnitude of the magnetic field at O is



- (a) zero (b) $\frac{\mu_0 I}{4\pi a}$ (c) $\frac{\mu_0 I}{2\pi a}$ (d) $\frac{\mu_0 I}{2\sqrt{2}\pi a}$

Sol: Sections AB and DE produce no field at O . Sections BC and EF produce equal fields at O , each $= \frac{\mu_0 I}{(4\pi a)}$

The magnetic of magnetic field at O is

$$B = B_{AB} + B_{BC} + B_{DE} + B_{EF}$$

$$= 0 + \frac{\mu_0 I}{4\pi a} + 0 + \frac{\mu_0 I}{4\pi a} = \frac{\mu_0 I}{2\pi a}$$

Ans: (c)

158. The moment of a magnet ($15 \text{ cm} \times 2 \text{ cm} \times 1 \text{ cm}$) is 1.2 A-m^2 . What is its intensity of magnetisation?

- (a) $4 \times 10^4 \text{ Am}^{-1}$ (b) $2 \times 10^{-4} \text{ Am}^{-1}$ (c) 10^4 Am^{-1} (d) None of these

Sol: Intensity of magnetisation

$$I_m = \frac{M}{V} = \frac{1.2}{(15 \times 2 \times 1)10^{-6}} = 4 \times 10^4 \text{ A m}^{-1}$$

Ans: (a)

159. Nucleus of an atom whose atomic mass is 24 consists of

- (a) 11 electrons, 11 protons and 13 neutrons (b) 11 electrons, 13 protons and 11 neutrons
(c) 11 protons and 13 neutrons (d) 11 protons and 13 electrons

Sol: Nucleus does not contain electron.

Ans: (c)

160. A tritium contains

- (a) 1 proton + 2 neutrons (b) 1 proton + 1 neutron
(c) 2 protons + 1 neutron (d) 2 protons + 2 neutrons

Sol: Triton is the nucleus of tritium atom.

$$\therefore \text{Triton} = {}_1H^3$$

It has one proton and 2 neutrons.

Ans: (a)

161. According to classical theory, the circular path of an electron in Rutherford atom model is

- (a) Spiral (b) Circular (c) Parabolic (d) Straight line

Sol: Spiral

Ans: (a)

162. As one considers orbits with higher values of n in a hydrogen atom, the electric potential energy of the atom

- (a) Decreases (b) Increases (c) Remains the same (d) Does not increase

Sol: Increases

Ans: (b)

163. Which of the following statements is true for hydrogen atom?

- (a) Angular momentum $\propto \frac{1}{n}$ (b) Linear momentum $\propto \frac{1}{n}$
(c) Radius $\propto \frac{1}{n}$ (d) Energy $\propto \frac{1}{n}$

Sol: Angular momentum

$$L = mvr = \frac{nh}{2\pi} \text{ or } mv = \frac{nh}{2\pi r}$$

Now, $r \propto n^2$

$$\therefore mv = p \propto \frac{nh}{2\pi \times n^2}$$

$$p \propto \frac{nh}{2\pi n} \text{ or } p \propto \frac{1}{n}$$

Energy, $E \propto \frac{1}{n^2}$

Ans: (b)

164. Doubly ionised helium atom and hydrogen ions are accelerated, from rest, through the same potential difference. The ratio of final velocities of helium and hydrogen is

- (a) $1:\sqrt{2}$ (b) $\sqrt{2}:1$ (c) $1:2$ (d) $2:1$

Sol: $qV = \frac{1}{2}mv^2$ or $v = \sqrt{\frac{2qV}{m}}$ i.e. $v \propto \sqrt{\frac{q}{m}}$

$$\therefore \frac{v_{He}}{v_H} = \sqrt{\frac{q_{He}}{q_H} \times \frac{m_H}{m_{He}}} = \sqrt{\frac{2e}{e} \times \frac{m}{4m}} = \frac{1}{\sqrt{2}}$$

Ans: (a)

165. What is conductivity of a semiconductor (in $\Omega^{-1}m^{-1}$) if electron density $= 5 \times 10^{12}/cm^3$ and hole density $= 8 \times 10^{13}/cm^3$? ($\mu_e = 2.3m^2 V^{-1}s^{-1}$, $\mu_h = 0.01m^2 V^{-1}s^{-1}$)

- (a) 5.634 (b) 1.968 (c) 3.421 (d) 8.964

Sol: Here, $\mu_e = 2.3m^2 V^{-1}s^{-1}$,

$$\mu_h = 0.01m^2 V^{-1}s^{-1}$$

$$n_e = 5 \times 10^{12}/cm^3 = 5 \times 10^{18}/m^3$$

$$n_h = 8 \times 10^{13}/cm^3 = 8 \times 10^{19}/m^3$$

$$\sigma = e(n_e\mu_e + n_h\mu_h)$$

$$= 1.6 \times 10^{-19} [(5 \times 10^{18}) \times 2.3 + (8 \times 10^{19}) \times 0.01]$$

$$= 1.6 \times 10^{-1} [11.5 + 0.8]$$

$$= 1.968 \Omega^{-1}m^{-1}$$

Ans: (b)

166. If c is the velocity of light, which of the following is correct?

- (a) $\mu_0\epsilon_0 = c$ (b) $\mu_0\epsilon_0 = c^2$ (c) $\mu_0\epsilon_0 = \frac{1}{c}$ (d) $\mu_0\epsilon_0 = \frac{1}{c^2}$

Sol: $c = \frac{1}{\sqrt{\mu_0\epsilon_0}}$ or $\mu_0\epsilon_0 = \frac{1}{c^2}$

Ans: (d)

167. Temperature coefficient of resistance of semiconductor is

- (a) Zero (b) Constant (c) Positive (d) Negative

Sol: The temperature coefficient of resistance of a semiconductor is negative. It means that resistance decrease with increase of temperature.

Ans: (d)

168. A potential barrier of 0.3 V exists across a $p-n$ junction. If the depletion region is $1\mu m$ wide, what is the intensity of electric field in this region?

- (a) $2 \times 10^5 V m^{-1}$ (b) $3 \times 10^5 V m^{-1}$ (c) $4 \times 10^5 V m^{-1}$ (d) $5 \times 10^5 V m^{-1}$

Sol: Electric field

$$E = \frac{V}{d} = \frac{0.3}{1 \times 10^{-6}} = 3 \times 10^5 \text{ V m}^{-1}$$

Ans: (b)

169. In the half wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be

- (a) 25 Hz (b) 50 Hz (c) 70.7 Hz (d) 100 Hz

Sol: In half wave rectifier, we get the output only in one half cycle of input a.c. therefore, the frequency of the ripple of the output is same as that of input a.c. i.e. 50 Hz

Ans: (b)

170. Ozone layer earth's atmosphere will not

- (a) Prevent infrared radiations from sun reaching earth
(b) Prevent infra-red radiations originated from earth from escaping earth's atmosphere
(c) Prevent ultraviolet rays from sun
(d) Reflect back radio waves

Sol: Ozone layer will absorb ultraviolet rays; reflect the infrared radiation and does not reflect back radio waves.

Ans: (d)

171. A circular coil of radius 0.1 m has 80 turns of wire. If the magnetic field through the coil increases from 0 to 2 tesla in 0.4 sec and the coil is connected to a 11 ohm resistor, what is the current (in A) flow through the resistor during the 0.4 sec?

- (a) (8/7) A (b) (7/8) A (c) 8 A (d) 7 A

$$\text{Sol: } |\varepsilon| = N \frac{d\phi}{dt} = NA \frac{dB}{dt}$$

$$= 80 \times \frac{22}{7} \times (0.1)^2 \left(\frac{2.0}{0.4} \right)$$

$$I = \frac{|\varepsilon|}{R} = 80 \times \frac{22}{7} \times \frac{(0.1)^2 \times 5}{11} = \frac{8}{7} \text{ A}$$

Ans: (a)

172. In a series LCR circuit the voltage across the resistance, capacitance and inductance is 10 V each. If the capacitance and inductance is short circuited. The voltage across the inductance will be

- (a) 10 V (b) $10\sqrt{2}$ V (c) $(10/\sqrt{2})$ V (d) 20 V

Sol: Here, $R = X_L = X_C$ ($\because$ voltage across them is same)

Total voltage in the circuit,

$$V = I[R^2 + (X_L - X_C)^2]^{1/2} = IR = 10 \text{ volt}$$

When capacitor is short circuited,

$$I' = \frac{10}{(R^2 + X_L^2)^{1/2}} = \frac{10}{\sqrt{2}R}$$

∴ Potential drop across inductance

$$= I' X_L = I' R = \frac{10}{\sqrt{2}} \text{ V}$$

Ans: (c)

173. If N is the number of turns in a coil, the value of self-inductance varies as

- (a) N^0 (b) N (c) N^2 (d) N^{-2}

Sol: Self Inductance = $\frac{\mu_0 N^2 A}{l}$

Ans: (c)

174. In an ac circuit, the reactance is equal to the resistance. The power factor of the circuit will be

- (a) 1 (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) zero

Sol: $\tan \phi = \frac{X}{R} = 1$

$$\phi = \tan^{-1}(1) = 45^\circ$$

$$\text{Power factor} = \cos \phi = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Ans: (c)

175. The core of a transformer is laminated to reduce

- (a) flux leakage (b) hysteresis (c) copper loss (d) eddy current

Sol: The core of a transformer is laminated to reduce eddy current.

Ans: (d)

176. The image formed by an objective of a compound microscope is

- (a) virtual and diminished (b) real and diminished
(c) real and enlarged (d) virtual and enlarged

Sol: The objective of compound microscope forms a real and enlarged image

Ans: (c)

177. A substance is behaving as convex lens in air and concave in water, then its refractive index is

- (a) greater than air but less than water (b) greater than both air and water
(c) smaller than air (d) almost equal to water

Sol: When a lens is immersed in a medium whose refractive index is greater than that of the lens, its nature changes. Here the lens changes its nature when immersed in water it means its refractive index is less than that of water.

Ans: (a)

178. The locus of all particles in a medium, vibrating in the same phase is called

- (a) Wavelet (b) fringe (c) wave front (d) none of these

Sol: Light is an electromagnetic wave of wavelength range $4000 \text{ \AA}$ to $7800 \text{ \AA}$

Ans: (c)

179. In a Young's double slit experiment, $d = 0.5\text{mm}$ and $D = 100\text{cm}$. It is found that 9th bright fringe is at a distance of 7.5 mm from the second dark fringe pattern. The wavelength of light used is (in Å)

- (a) $\frac{2500}{7}$ (b) 2500 (c) 5000 (d) $\frac{5000}{7}$

Sol: For bright fringes, $x = \frac{n\lambda D}{d}$

Where $n = 0, 1, 2, 3, \dots$

For dark fringes, $x = \frac{(2n-1)\lambda D}{2d}$

where $n = 1, 2, 3, \dots$

As per question

$$\frac{9\lambda D}{d} - \frac{3\lambda D}{2d} = 7.5 \times 10^{-3} \text{ m}$$

$$\text{or } \frac{15\lambda D}{2d} = 7.5 \times 10^{-3}$$

$$\text{or } \lambda = \frac{2 \times 7.5 \times 10^{-3} \text{ m} \times d}{15D}$$

Substituting the given values, we get

$$\lambda = \frac{2 \times 7.5 \times 10^{-3} \text{ m} \times 0.5 \times 10^{-3} \text{ m}}{15 \times 1 \text{ m}}$$

$$= 0.5 \times 10^{-6} \text{ m} = 5000 \text{ Å}$$

Ans: (c)

180. The focal length of a biconvex lens of refractive index 1.5 is 0.06 m. Radii of curvature are in the ratio 1: 2. Then radii of curvature of two lens surfaces are

- (a) 0.045 m, 0.09 m (b) 0.09 m, 0.18 m (c) 0.04 m, 0.08 m (d) 0.06 m, 0.12 m

Sol: According to lens maker's formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

For biconvex lens $R_1 = +R_1$; $R_2 = -R_2$

$$\frac{1}{0.06} = (1.5 - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right); \quad \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{0.06 \times 0.5} = \frac{100}{3} \quad \dots (i)$$

$$\text{Now, } \frac{R_1}{R_2} = \frac{1}{2} \text{ or } R_2 = 2R_1$$

On substituting it in eq (i), we get $\frac{3}{2R_1} = \frac{100}{3}$

$$R_1 = \frac{9}{200} = 0.045 \text{ m}$$

$$R_2 = 2R_1 = 2 \times 0.045 \text{ m} = 0.09 \text{ m}$$

Ans: (a)

Key Answers:

1. a	2. c	3. b	4. b	5. d	6. b	7. b	8. c	9. a	10. d
11. a	12. c	13. c	14. d	15. b	16. a	17. b	18. b	19. a	20. d
21. a	22. d	23. b	24. d	25. a	26. d	27. c	28. a	29. c	30. s
31. a	32. c	33. c	34. c	35. a	36. b	37. c	38. d	39. c	40. d
41. c	42. c	43. a	44. c	45. c	46. c	47. b	48. a	49. d	50. c
51. c	52. d	53. a	54. c	55. c	56. b	57. b	58. d	59. b	60. a
61. d	62. b	63. d	64. b	65. a	66. d	67. c	68. c	69. a	70. d
71. c	72. c	73. d	74. b	75. c	76. c	77. c	78. b	79. b	80. c
81. a	82. d	83. b	84. a	85. c	86. c	87. c	88. c	89. a	90. b
91. d	92. b	93. c	94. a	95. b	96. b	97. c	98. b	99. a	100.c
101.d	102.d	103.d	104.b	105.d	106.b	107.c	108.d	109.a	110.b
111.b	112.b	113.c	114.b	115.b	116.b	117.d	118.d	119.c	120.b
121.a	122.b	123.c	124.b	125.a	126.c	127.c	128.b	129.b	130.a
131.a	132.c	133.c	134.c	135.c	136.a	137.d	138.b	139.d	140.b
141.c	142.b	143.a	144.a	145.a	146.b	147.b	148.d	149.b	150.c
151.b	152.a	153.b	154.c	155.c	156.c	157.c	158.a	159.c	160.a
161.a	162.b	163.b	164.a	165.b	166.d	167.d	168.b	169.b	170.d
171.a	172.c	173.c	174.c	175.d	176.c	177.a	178.c	179.c	180.a